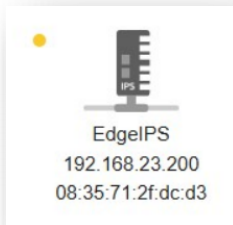
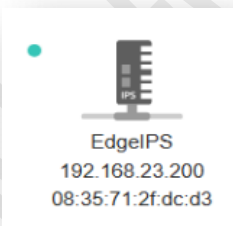
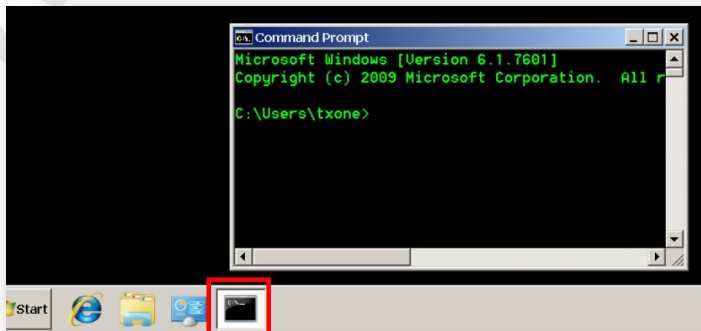
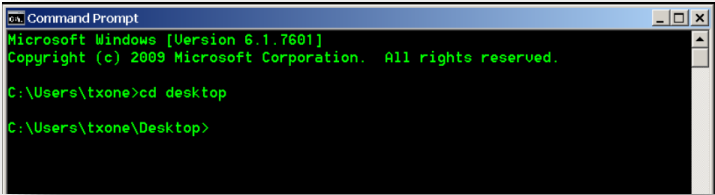
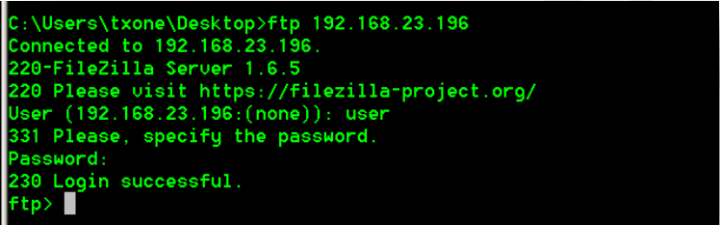
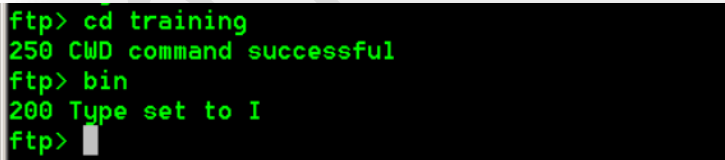
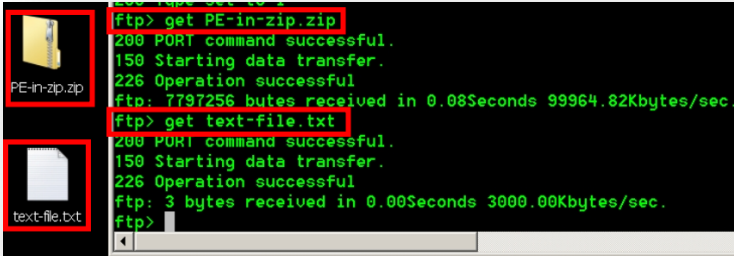
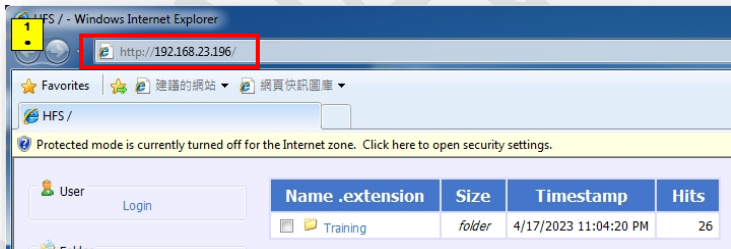
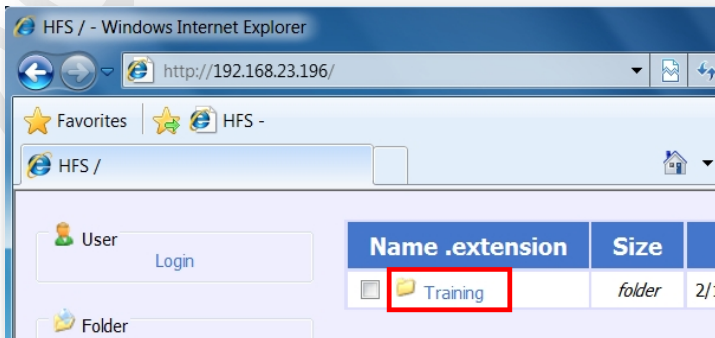


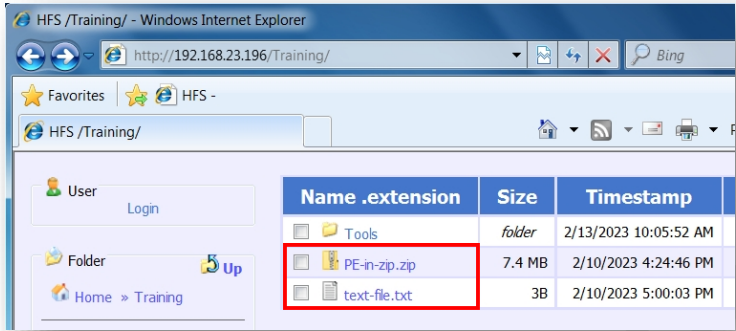
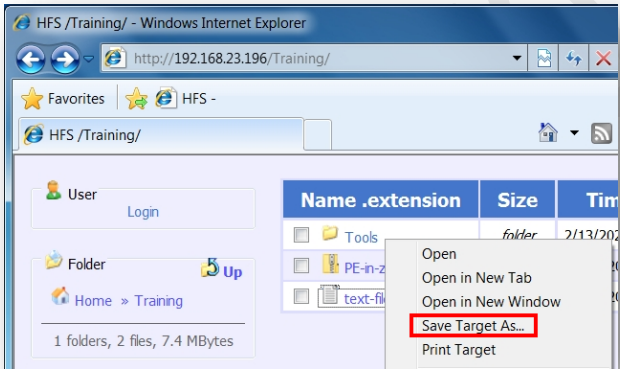
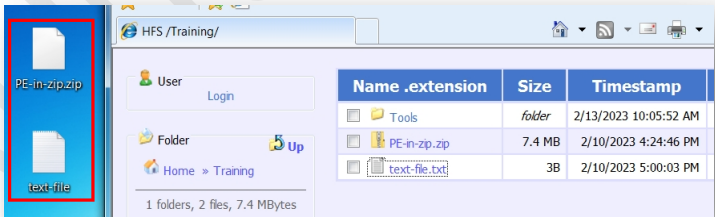
Items	Descriptions
	4. Select the device group “FAB12-10-27”. There should now be one EdgeIPS device in it. There may still be a yellow light next to the EdgeIPS device icon. This means EdgeOne is now syncing group policy settings with EdgeIPS.  EdgeIPS 192.168.23.200 08:35:71:2f:dc:d3 5. The indicator will change to green when the EdgeOne group policy settings are synced with the EdgeIPS device.  EdgeIPS 192.168.23.200 08:35:71:2f:dc:d3
Result and Comments	The Edge group “ FAB12-10-27 ” created and EdgeIPS moved into it.

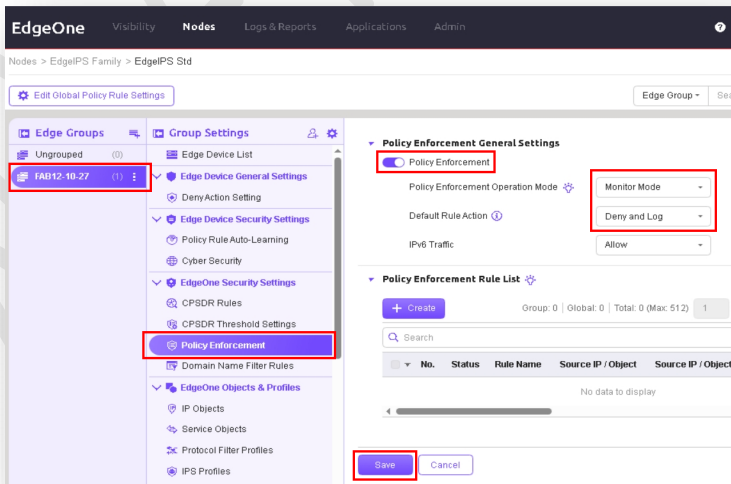
Lab7: Policy Enforcement Configuration

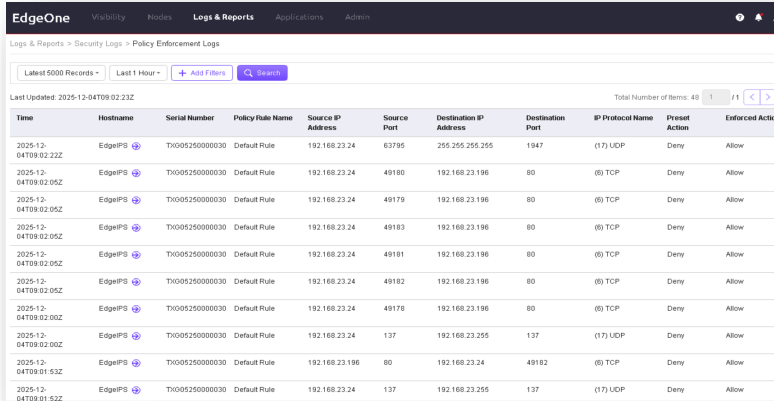
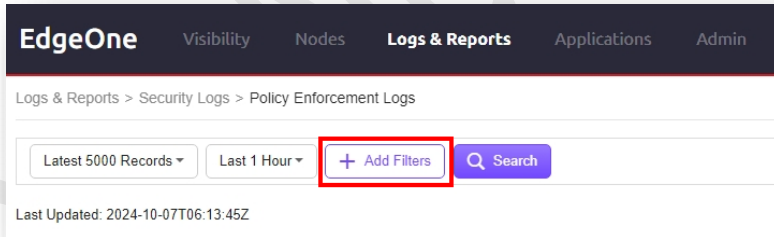
Items	Descriptions
Application target	Host A and B
Steps	Things to check before starting this Lab exercise. 1. The LabConsole VM (which must run on Host A) has a hosted FTP server and a web server. Those will be used to verify the EdgeIPS policy enforcement. 2. In the next steps, you will “Download files over FTP” or “Download files over HTTP”. Both activities are done from the Win7 virtual machine, which should run on Host B. Use the following credentials. User: txone Password: 12345678 3. Let’s first test if this works without the EdgeIPS (the EdgeIPS should be powered off) Download files over FTP. This is a “Standard Operating Procedure” (SOP). It can be seen as a “subroutine” in this lab. In the next steps, you will be asked to do this SOP a few times, but every time with different policy settings in the EdgeIPS. 1. On the VM Win7, which should run on Host B, open cmd from task bar 

Items	Descriptions
	2. Enter "CD \users\txone\desktop"  3. Enter ftp 192.168.23.196 to Log in to the FTP server, using the following credentials User: user Password: 12345678  4. Enter "cd training" to go to the "training" directory. 5. Enter "bin" to switch to binary mode.  6. Enter "get PE-in-zip.zip" and "get text-file.txt" and check if the files have download successfully or not. If the download was successful, you should see the files on your desktop. 

Items	Descriptions
	7. Enter “quit” to exit the FTP client 8. Enter “exit” to exit the CMD 9. Delete the PE-in-zip.zip and text-file.zip files from the desktop. Download files over HTTP. 1. On the VM Win7, which should run on Host B, Launch the Internet Explorer (IE) browser from the taskbar.  2. IE will access the HFS web server automatically (Which runs on Host A). Alternatively, manually go to: http://192.168.23.196  3. Click on the “Training” web folder. 

Items	Descriptions
	4. Download PE-in-zip.zip, text-file.txt and store the files on desktop.  5. To download the text file, right click it and choose “Save Target As...”  6. Check if the files successfully downloaded to the desktop or not.  7. Close the IE browser. 8. Delete the PE-in-zip.zip and text-file.txt from the desktop. 9. This step was simply to observe the system's behavior without EdgeIPS in the connection.

Items	Descriptions
	Monitor Mode versus Prevention Mode Objective : 1. Demonstrate FTP and HTTP access are functional under Monitor Mode. 2. Demonstrate that, after switching to Prevention Mode, access is blocked (Deny All) Steps: 1. Power on the EdgeIPS 2. Log in to the EdgeOne Web UI, go to https://192.168.23.21 or click the “EdgeOne” bookmark in Chrome. 3. Go to Node > EdgeIPS Family > EdgeIPS Std 4. Select the Edge Group “FAB12-10-27” 5. Select “Policy Enforcement” 6. Enable “Policy Enforcement” 7. Change the Policy Enforcement Operation Mode to “Monitor Mode” 8. Set the Policy Enforcement Default Rule Action to “Deny and Log” 9. Click Save  10. Execute the “Download files over FTP” SOP. The files should be successfully downloaded by FTP.

Items	Descriptions
	11. Execute the “Download files over http” SOP. The files should be successfully downloaded over HTTP. 12. Go to Logs & Reports > Security Logs > Policy Enforcement Logs to check the logs.  13. You can apply event filters if you see too much “noise”. Click on the Add Filter(s) button.  14. Entry 1st filter - Field: Source IP Address - Operator: is - Value: 192.168.23.24 15. Entry 2nd filter - Field: IP Protocol Name - Operator: is - Value: (6) TCP

Items	Descriptions
-------	--------------

Add Filters

×

Add up to 10 filters to search for specific logs. In each filter you can include up to 10 search values.

Field

Source IP Address

Operator

is

Value

192.168.23.24

Field

IP Protocol Name

Operator

is

Value

(6) TCP

16. With the filters applied, the logs are easier to check.

Policy Rule Name	Source IP Address	Source Port	Destination IP Address	Destination Port	IP Protocol Name	Preset Action	Enforced Action
Default Rule	192.168.23.24	49178	192.168.23.196	80	(6) TCP	Deny	Allow
Default Rule	192.168.23.24	49179	192.168.23.196	80	(6) TCP	Deny	Allow
Default Rule	192.168.23.24	49180	192.168.23.196	80	(6) TCP	Deny	Allow
Default Rule	192.168.23.24	49183	192.168.23.196	80	(6) TCP	Deny	Allow
Default Rule	192.168.23.24	49181	192.168.23.196	80	(6) TCP	Deny	Allow
Default Rule	192.168.23.24	49182	192.168.23.196	80	(6) TCP	Deny	Allow
Default Rule	192.168.23.24	49168	192.168.23.196	21	(6) TCP	Deny	Allow
Default Rule	192.168.23.24	49168	192.168.23.196	21	(6) TCP	Deny	Allow
Default Rule	192.168.23.24	49168	192.168.23.196	21	(6) TCP	Deny	Allow

17. Set Policy Enforcement to “Prevention Mode”

▼ Policy Enforcement General Settings

Policy Enforcement

Policy Enforcement Operation Mode

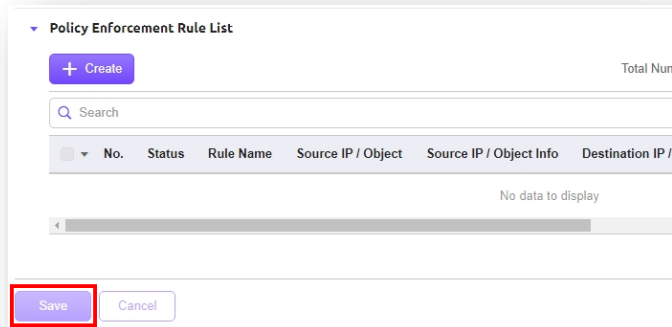
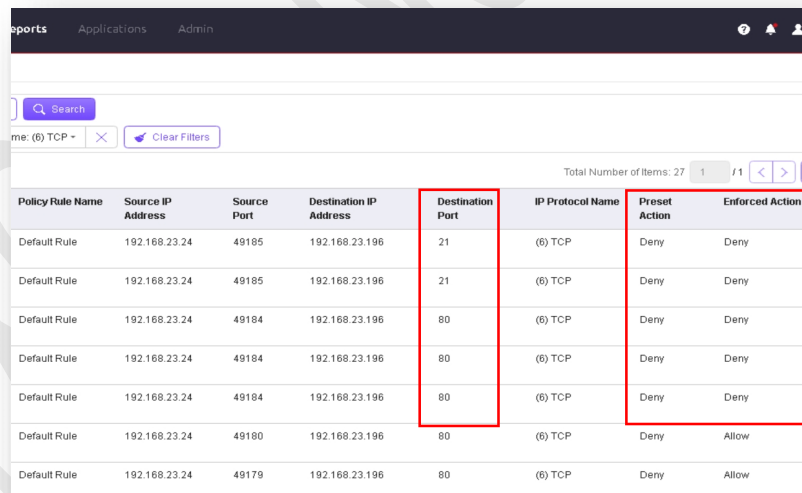
Default Rule Action

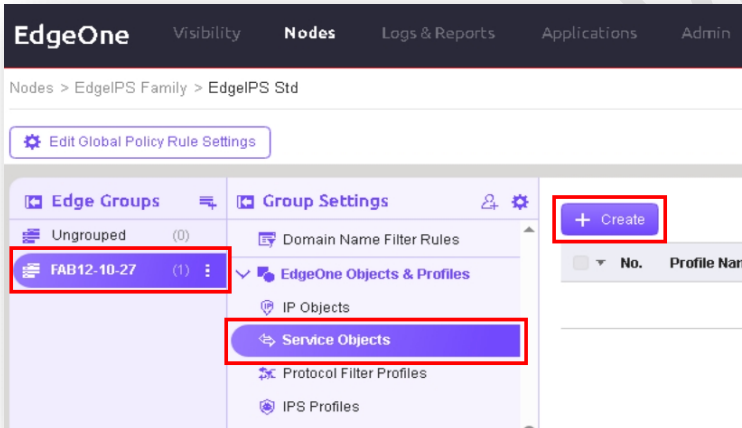
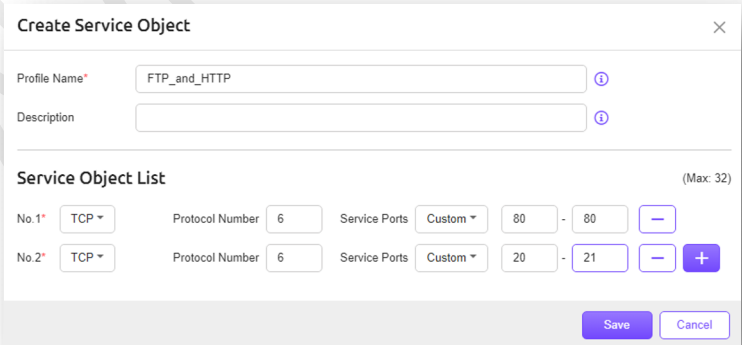
IPv6 Traffic

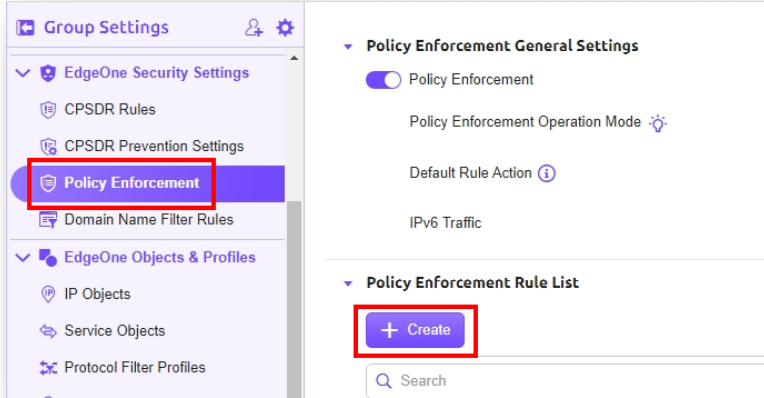
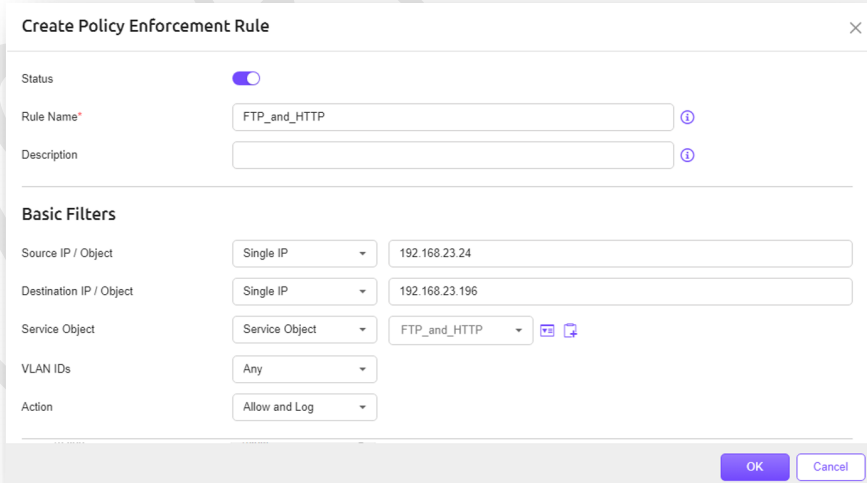
Prevention Mode

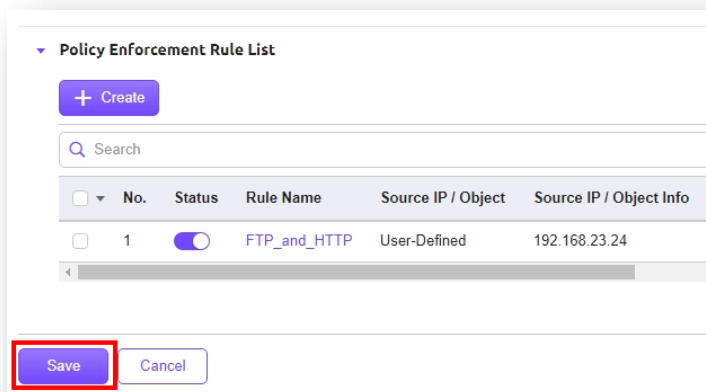
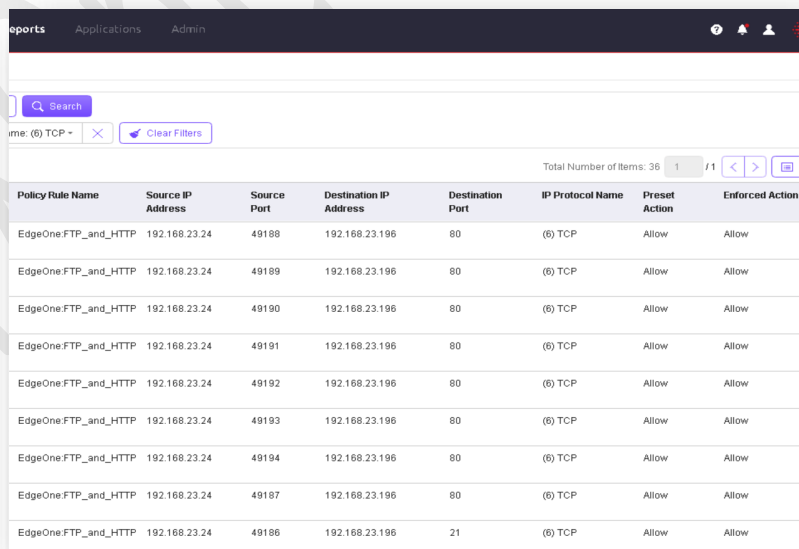
Deny and Log

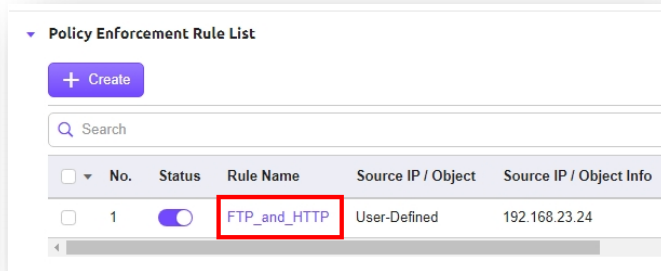
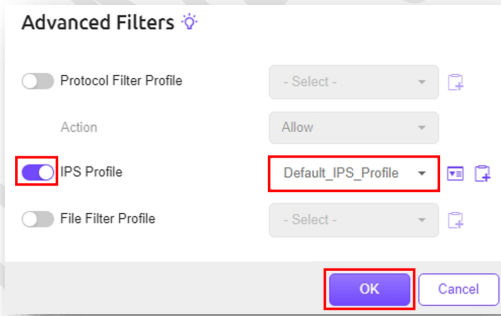
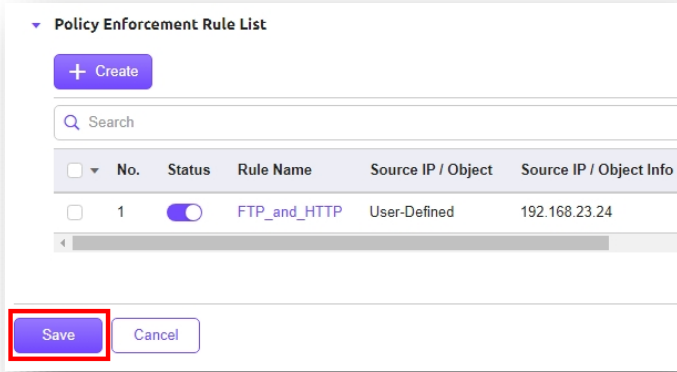
Allow

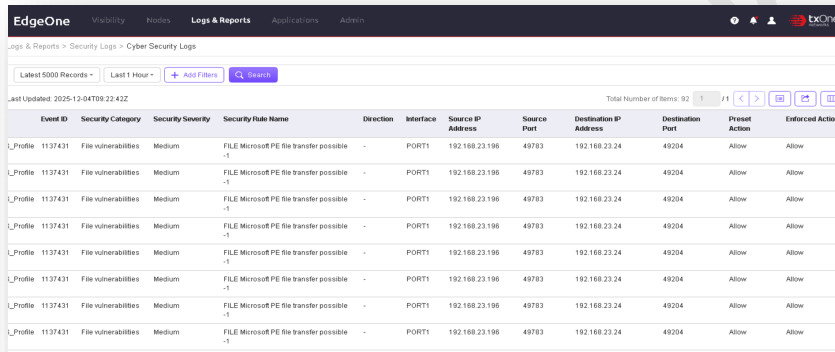
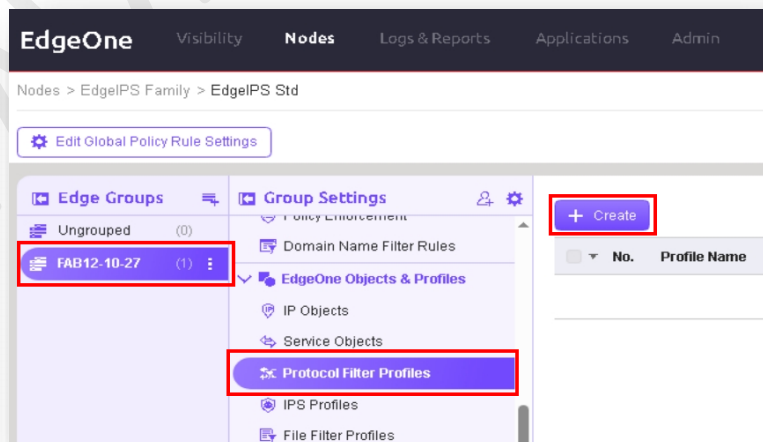
Items	Descriptions
	18. Remember to click Save to save the settings.  19. Execute the “Download files over FTP” SOP. The file download over FTP should Fail. 20. Execute the “Download files over http” SOP. The file download over http should Fail. 21. Go to Logs & Reports > Security Logs > Policy Enforcement Logs to check the logs. Hint: You can apply filter rules to filter the logs to find the result easier. 

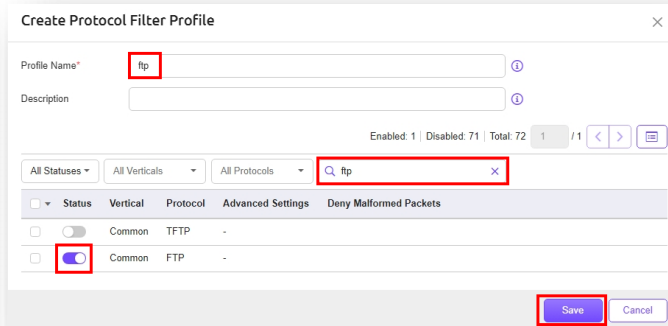
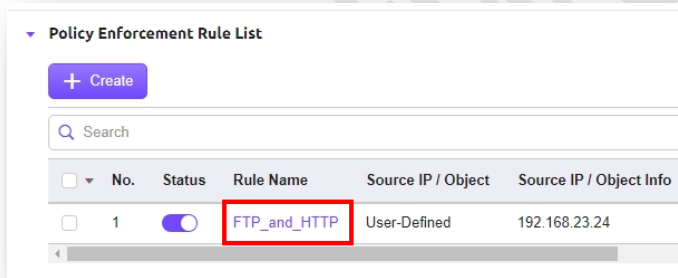
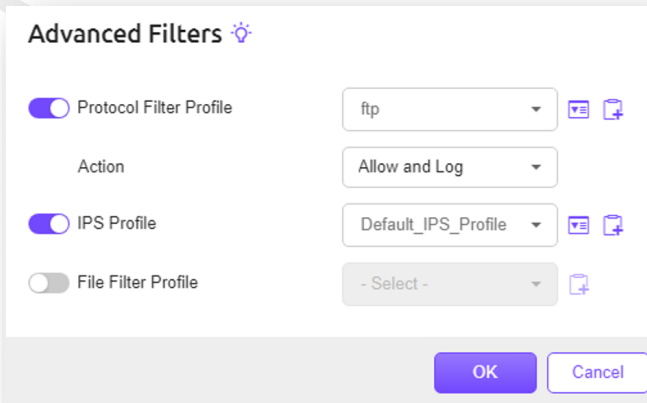
Items	Descriptions
	Configure a “Basic filter rule” to allow ftp and http access. Create a “Service Object” for ftp and http - Go to Node > EdgeIPS Family > EdgeIPS Std - Select the Edge Group “FAB12-10-27” - Select the “Service Object” - Click + Create.  - Edit the Profile Name: FTP_and_HTTP. - No1: Protocol: TCP, Service Port: Custom, 80 ~ 80. - No2: Protocol: TCP, Service Port: Custom, 20 ~ 21. - Click Save. 

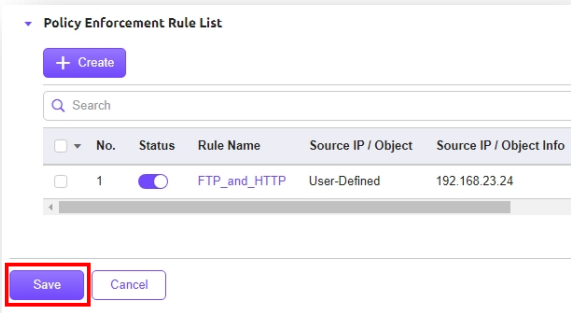
Items	Descriptions
	7. Select Policy Enforcement 8. Click + Create.  9. Rule Name: FTP_and_HTTP - Source IP / Single IP: 192.168.23.24 - Destination IP / Single IP: 192.168.23.196 - Service Object: Select "FTP_and_HTTP" - Action: Accept and Log 10. Click OK 

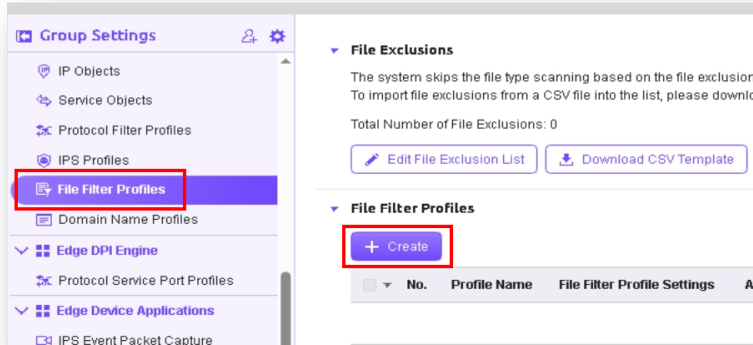
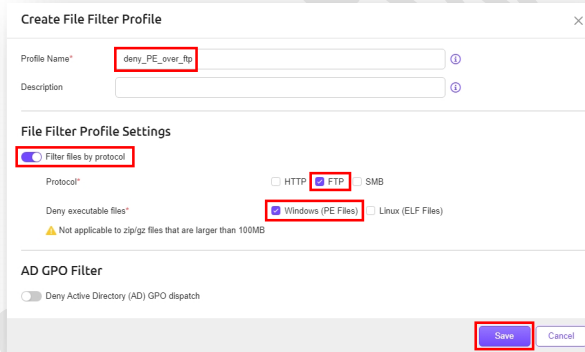
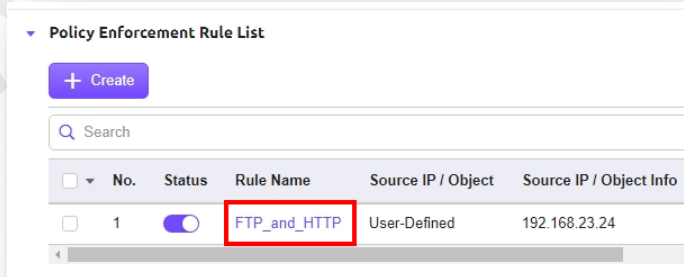
Items	Descriptions
	11. Click Save to apply the policy.  12. Execute the “Download files over FTP” SOP. The file download over FTP should be successful. 13. Execute the “Download files over http” SOP. The file download over http should be successful. 14. Go to Logs & Reports > Security Logs > Policy Enforcement Logs to check the logs. Tips: You can add a search filter Rule Name:FTP_and_HTTP to make it easier to check the logs. 

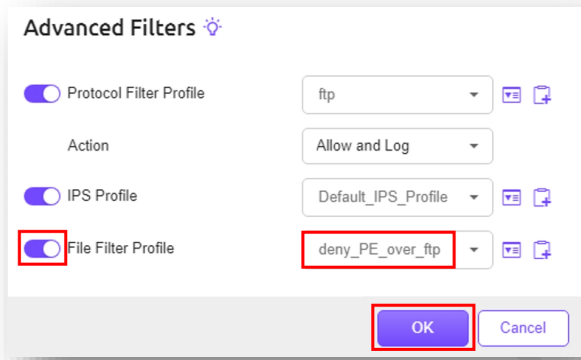
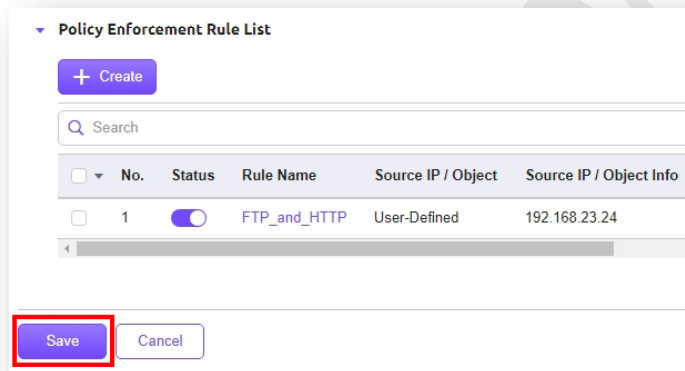
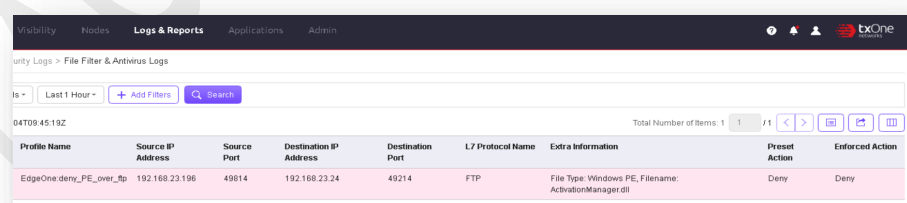
Items	Descriptions
	Configure Advanced Filter – IPS Profile to detect suspicious / malicious behavior. - Go to Nodes > EdgeIPS Family > EdgeIPS Std - Select the Edge Group: FAB12-10-27 - Select Policy Enforcement - Click on the Rule Name: “FTP_and_HTTP”  The screenshot shows the 'Policy Enforcement Rule List' interface. It includes a '+ Create' button, a search bar, and a table with columns: No., Status, Rule Name, Source IP / Object, and Source IP / Object Info. The first row shows rule number 1, status 'On', rule name 'FTP_and_HTTP' (highlighted with a red box), source 'User-Defined', and IP '192.168.23.24'. Enable the IPS Profile in Advanced Filter, choose the “Default_IPS_Profile” and Click OK.  The screenshot shows the 'Advanced Filters' configuration dialog. It has three sections: 'Protocol Filter Profile' (disabled), 'IPS Profile' (enabled with a red box around the toggle), and 'File Filter Profile' (disabled). The 'IPS Profile' dropdown is set to 'Default_IPS_Profile' (highlighted with a red box). The 'Action' is set to 'Allow'. At the bottom, there are 'OK' and 'Cancel' buttons, with 'OK' highlighted by a red box. - Click Save to save the policy.  The screenshot shows the 'Policy Enforcement Rule List' interface again. At the bottom, there are 'Save' and 'Cancel' buttons, with the 'Save' button highlighted by a red box.

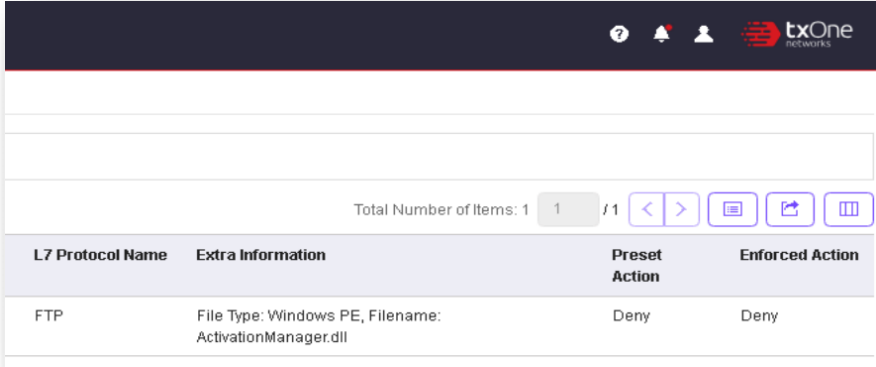
Items	Descriptions
	- Execute the “Download files over FTP” SOP. The file download over FTP should be successful. - Execute the “Download files over http” SOP. The file download over FTP should be successful. - Those download activities will trigger the suspicious PE file transfer cyber security events. - Check the Logs & Reports > Security Logs > Cyber Security Logs
	 The screenshot shows the EdgeOne interface with the 'Logs & Reports' tab selected. Under 'Security Logs', the 'Cyber Security Logs' are displayed. The table lists events with columns for Event ID, Security Category, Security Severity, Security Rule Name, Direction, Interface, Source IP Address, Source Port, Destination IP Address, Destination Port, Preset Action, and Enforced Action. The events are all categorized as 'File vulnerabilities' with a 'Medium' severity, triggered by the rule 'FILE Microsoft PE file transfer possible -1'. The source IP is 192.168.23.196 and the destination IP is 192.168.23.24. The destination port is 49204. The preset and enforced actions are both 'Allow'.
	Configure Advanced Filter – Protocol filter Profile to allow ftp protocol only - Go to Nodes > EdgeIPS Family > EdgeIPS Std - Select the Edge Group “FAB12-10-27”. - Choose the “Protocol Filter Profiles”. - Click on the “+ Create” button.
	 The screenshot shows the EdgeOne 'Nodes' configuration page for 'EdgeIPS Family > EdgeIPS Std'. The 'Edge Groups' list on the left includes 'Ungrouped (0)' and 'FAB12-10-27 (1)'. The 'FAB12-10-27' group is selected. On the right, the 'Group Settings' panel shows 'Policy Enforcement' and 'Domain Name Filter Rules'. The 'EdgeOne Objects & Profiles' section is expanded, showing 'IP Objects', 'Service Objects', 'Protocol Filter Profiles', 'IPS Profiles', and 'File Filter Profiles'. The 'Protocol Filter Profiles' option is highlighted. A '+ Create' button is visible in the top right corner of the settings panel.

Items	Descriptions
	5. Profile Name: ftp. 6. Search: ftp, Enable FTP protocol, and click Save.  7. Add the Protocol Filter Profile to Policy Enforcement. Click on the Policy Enforcement Rule Name: "FTP_and_HTTP".  8. Enable the Protocol Filter Profile. 9. Choose "ftp" profile. 10. Set Action to "Accept and Log" 11. Click OK. 

Items	Descriptions																																																																								
	12. Click Save to enforce the policy.  13. Execute the “Download files over FTP” SOP. The file download over FTP should be successful. 14. Execute the “Download files over http SOP”. The file download over HTTP should fail because only FTP is allowed to pass through. 15. Check the Logs & Reports > Security Logs > Protocol Filter Logs. <table><tr><th>Source IP Address</th><th>Source Port</th><th>Destination IP Address</th><th>Destination Port</th><th>IP Protocol Name</th><th>L7 Protocol Name</th><th>Preset Action</th><th>Enforced Action</th></tr><tr><td>192.168.23.196</td><td>80</td><td>192.168.23.24</td><td>49206</td><td>(6) TCP</td><td>HTTP</td><td>Deny</td><td>Deny</td></tr><tr><td>192.168.23.24</td><td>49211</td><td>192.168.23.196</td><td>80</td><td>(6) TCP</td><td>HTTP</td><td>Deny</td><td>Deny</td></tr><tr><td>192.168.23.24</td><td>49211</td><td>192.168.23.196</td><td>80</td><td>(6) TCP</td><td>HTTP</td><td>Deny</td><td>Deny</td></tr><tr><td>192.168.23.196</td><td>80</td><td>192.168.23.24</td><td>49206</td><td>(6) TCP</td><td>HTTP</td><td>Deny</td><td>Deny</td></tr><tr><td>192.168.23.196</td><td>80</td><td>192.168.23.24</td><td>49206</td><td>(6) TCP</td><td>HTTP</td><td>Deny</td><td>Deny</td></tr><tr><td>192.168.23.24</td><td>49208</td><td>192.168.23.196</td><td>21</td><td>(6) TCP</td><td>FTP</td><td>Allow</td><td>Allow</td></tr><tr><td>192.168.23.196</td><td>49795</td><td>192.168.23.24</td><td>49210</td><td>(6) TCP</td><td>FTP</td><td>Allow</td><td>Allow</td></tr><tr><td>192.168.23.24</td><td>49208</td><td>192.168.23.196</td><td>21</td><td>(6) TCP</td><td>FTP</td><td>Allow</td><td>Allow</td></tr></table> Configure Advanced Filter – File Filter Profile to deny PE file transfer over ftp 1. Go to Nodes > EdgeIPS Family > EdgeIPS Std 2. Select the Edge Group “FAB12-10-27”. 3. Select the “File Filter Profiles” and click + Create.	Source IP Address	Source Port	Destination IP Address	Destination Port	IP Protocol Name	L7 Protocol Name	Preset Action	Enforced Action	192.168.23.196	80	192.168.23.24	49206	(6) TCP	HTTP	Deny	Deny	192.168.23.24	49211	192.168.23.196	80	(6) TCP	HTTP	Deny	Deny	192.168.23.24	49211	192.168.23.196	80	(6) TCP	HTTP	Deny	Deny	192.168.23.196	80	192.168.23.24	49206	(6) TCP	HTTP	Deny	Deny	192.168.23.196	80	192.168.23.24	49206	(6) TCP	HTTP	Deny	Deny	192.168.23.24	49208	192.168.23.196	21	(6) TCP	FTP	Allow	Allow	192.168.23.196	49795	192.168.23.24	49210	(6) TCP	FTP	Allow	Allow	192.168.23.24	49208	192.168.23.196	21	(6) TCP	FTP	Allow	Allow
Source IP Address	Source Port	Destination IP Address	Destination Port	IP Protocol Name	L7 Protocol Name	Preset Action	Enforced Action																																																																		
192.168.23.196	80	192.168.23.24	49206	(6) TCP	HTTP	Deny	Deny																																																																		
192.168.23.24	49211	192.168.23.196	80	(6) TCP	HTTP	Deny	Deny																																																																		
192.168.23.24	49211	192.168.23.196	80	(6) TCP	HTTP	Deny	Deny																																																																		
192.168.23.196	80	192.168.23.24	49206	(6) TCP	HTTP	Deny	Deny																																																																		
192.168.23.196	80	192.168.23.24	49206	(6) TCP	HTTP	Deny	Deny																																																																		
192.168.23.24	49208	192.168.23.196	21	(6) TCP	FTP	Allow	Allow																																																																		
192.168.23.196	49795	192.168.23.24	49210	(6) TCP	FTP	Allow	Allow																																																																		
192.168.23.24	49208	192.168.23.196	21	(6) TCP	FTP	Allow	Allow																																																																		

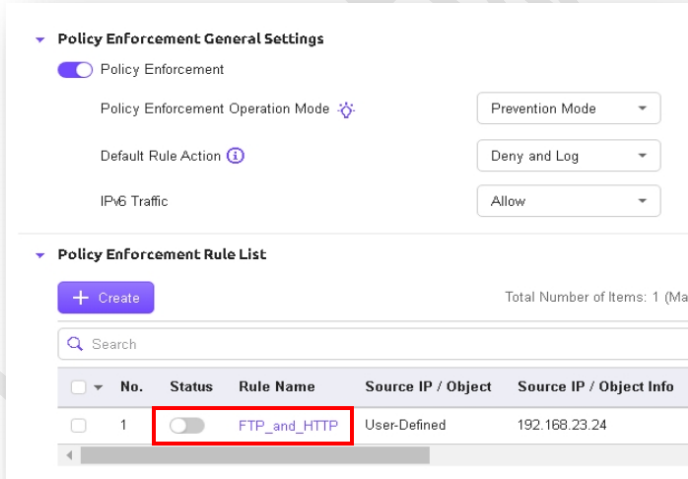
Items	Descriptions
	 - Profile Name: “deny_PE_over_ftp”. - Enable the “File Filter by Protocol”. - Check the Protocol: FTP. - Deny Executable File(s): Windows (PE Files).  - Add the File Filter Profile to the Policy Enforcement. Click the Policy Enforcement Rule Name: FTP_and_HTTP.  - Enable the File Filter Profile and choose the “deny_PE_over_ftp” profile. - Click OK

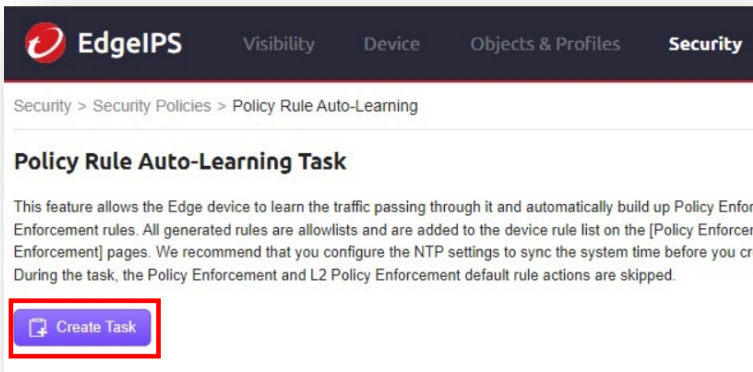
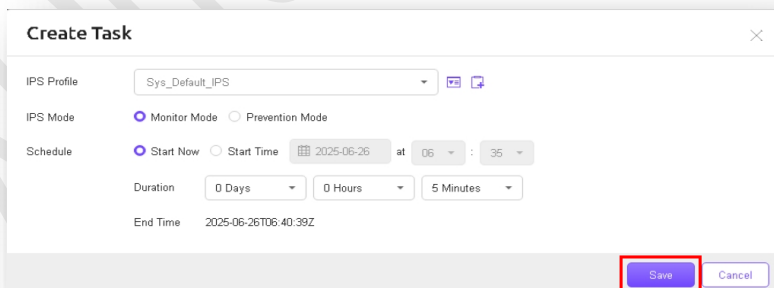
Items	Descriptions
	 11. Click Save to enforce the policy.  12. Execute the “Download files over FTP SOP”. Only the download of the text file over FTP should be successful. The download of the PE file will fail with a “Connection aborted” message. 13. Go to Logs & Reports > Security Logs > File Filter & Antivirus Logs. 

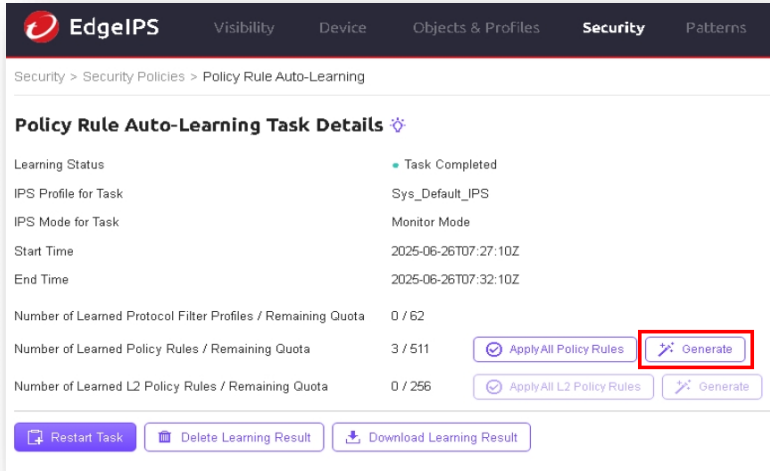
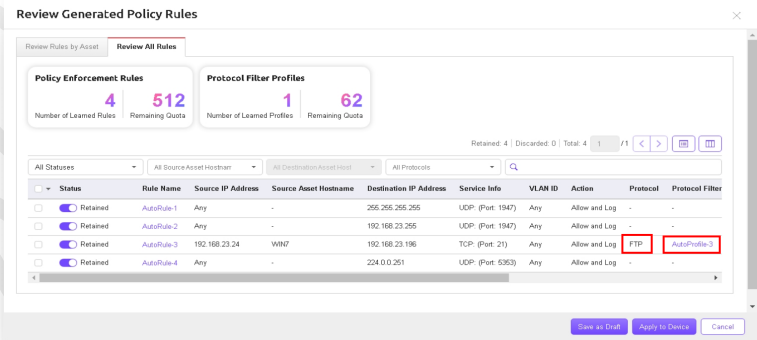
Items	Descriptions
	 Also check the following logs. Could you explain the result of logs? 14. Policy Enforcement Logs 15. Cyber Security logs 16. Protocol Filter Logs
Result and Comments	EdgeIPS “Basic” and “Advanced” filter of Policy Enforcement Rule tested

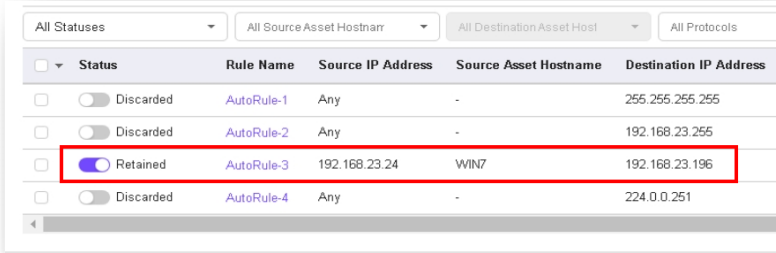
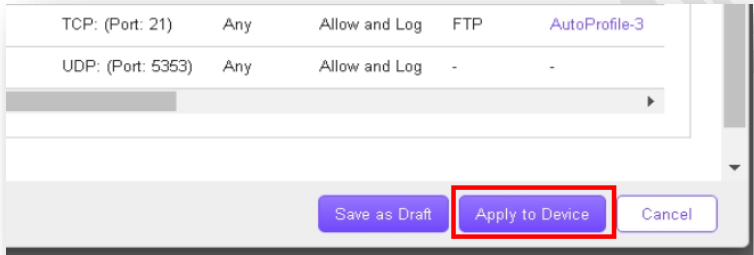
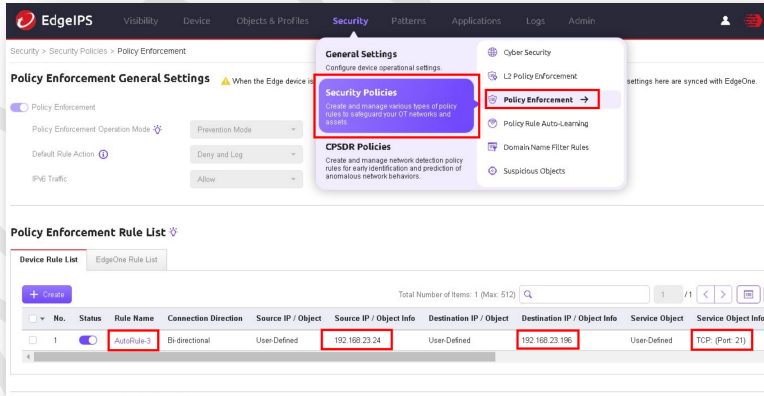
Lab8: EdgeIPS Policy Rule Auto-Learning

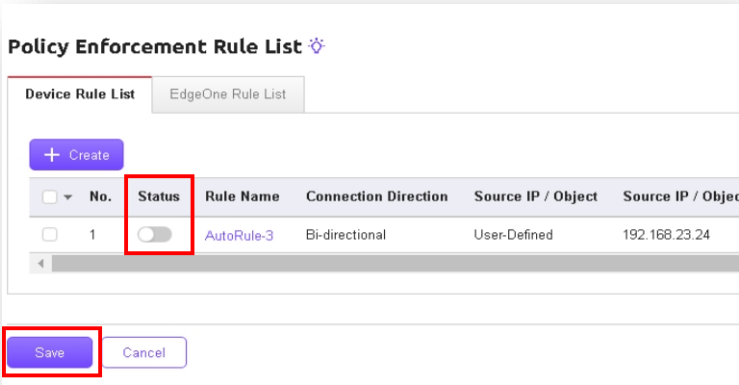
Table 11 EdgeIPS Policy Rule Auto-Learning

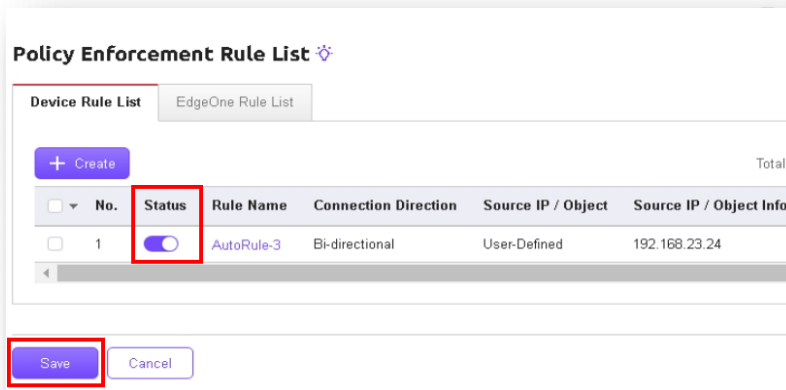
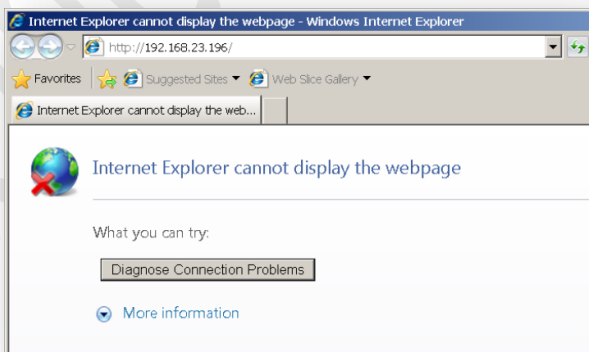
Items	Descriptions
Application Target	Host A & B
Steps	In this lab, we will enable EdgeIPS Policy Rule Auto-Learning to learn the behavior of allowing only the FTP protocol to pass through. 1. Logon to the EdgeOne UI 2. Go to the Nodes > EdgeIPS Family > EdgeIPS Std 3. Select the Edge Group FAB12-10-27 > Policy Enforcement 4. Disable the rule FTP_and_HTTP 

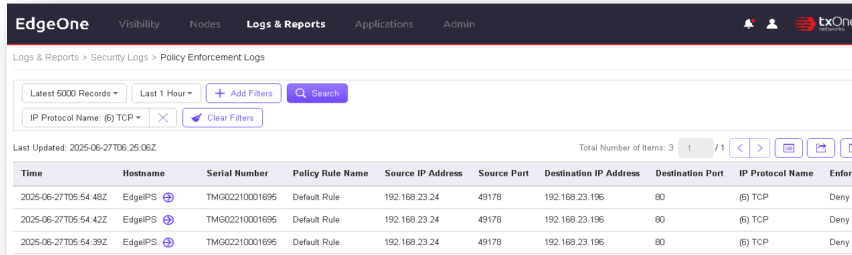
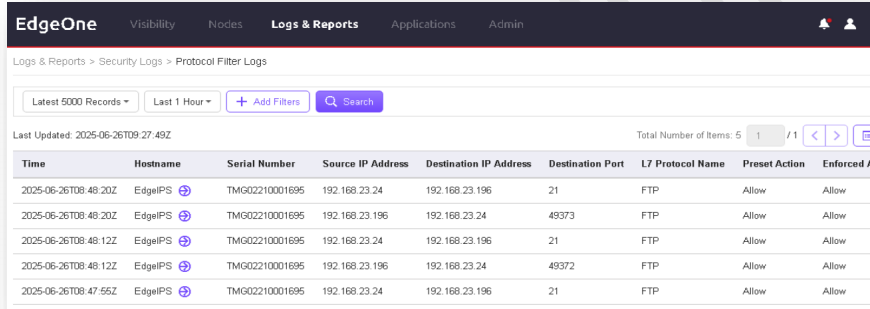
Items	Descriptions
	Enable EdgeIPS Policy Rule Auto-Learning to learn the Download files over FTP activities from the network traffic - Navigate to the EdgeIPS web UI, https://192.168.23.200 - Go to Security > Security Policy > Policy Rule Auto-Learning - Click on the Create Task button  - Configure the following settings and click Save. - IPS Profile: "Sys_Default_IPS" - IPS Mode: "Monitor Mode" - Execution Time: "Start Now" - Duration: 5 minutes  - Execute the "Download files over FTP" SOP. The files should be successfully downloaded by FTP.

Items	Descriptions
	10. Wait for the Auto learning task to complete and click the on the “Generate” button.  11. Click the “Review All Rules” tab and find the FTP access behaviors. - Source IP Address: 192.168.23.24 - Destination IP Address: 192.168.23.196 - Service Info: TCP Port 21 - Protocol: FTP 12. Click on the Protocol Filter Profile “AutoProfile-x” 

Items	Descriptions
	13. Discard all the rules that are not related to FTP.  14. Click on “Apply to Device”  15. Go to Security > Security Policies > Policy Enforcement, check the tab Device Rule List. A new rule Allow FTP access should have been created. 

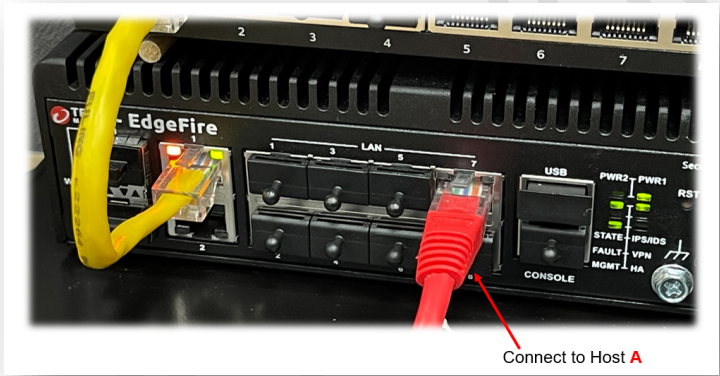
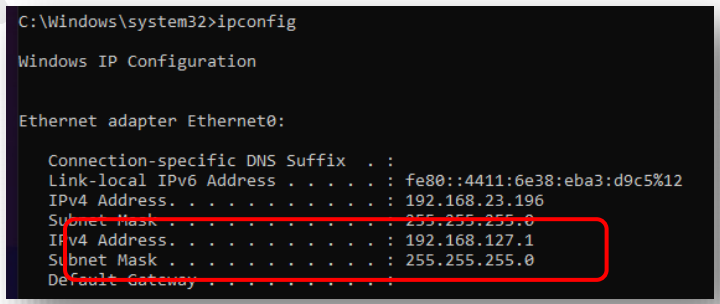
Items	Descriptions
	Verify HTTP access before enforcing the FTP access only 1. Logon to the EdgeIPS device web UI 2. Disable the Allow FTP access rule by clicking on the Status button. 3. Click Save.  4. Execute the “Download files over http” SOP. The files should fail to download over HTTP. 5. Execute the “Download files over ftp” SOP. The files should fail to download over FTP.

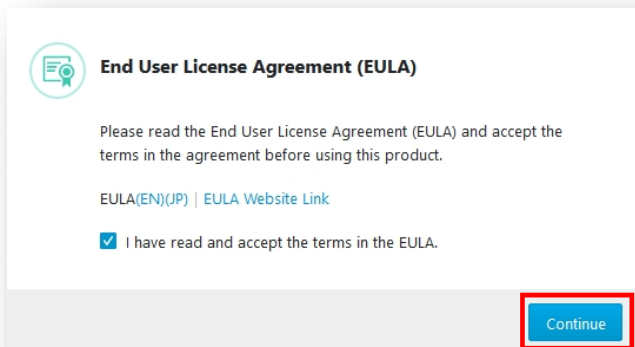
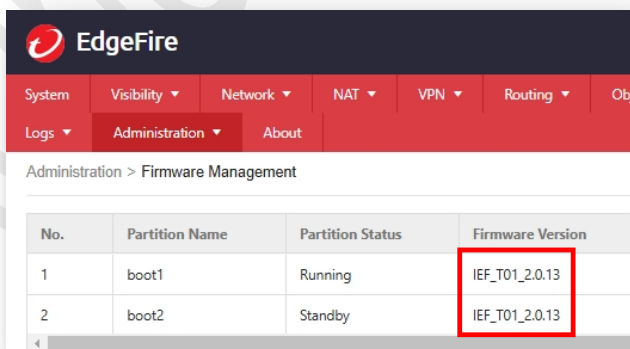
Items	Descriptions
	Enable EdgeIPS Policy Enforcement to block the unauthorized Download files over HTTP behavior from device rule list - Go to the EdgeIPS device web UI - Go to Security > Security Policies > Policy Enforcement. Enable the Rule that was learned from the traffic to allow only FTP Protocol to pass through the EdgeIPS.  - Execute the “Download files over ftp” SOP. The files should successfully download over FTP - Execute the “Download files over http” SOP. The files should fail to download over HTTP. 

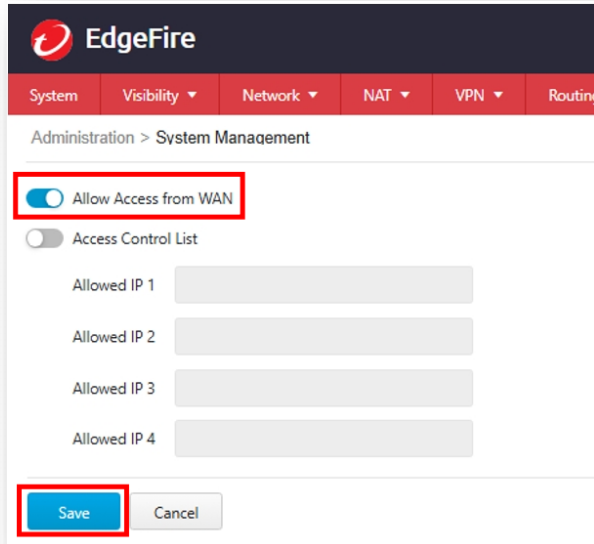
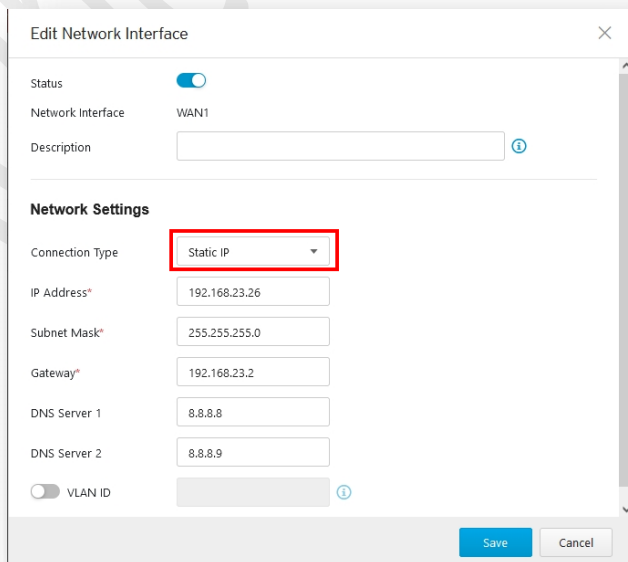
Items	Descriptions
	5. Go to Log & Reports > Security Logs > Policy Enforcement Logs. The HTTP traffic is denied by EdgeIPS.  6. Go to Logs & Reports > Security Logs > Protocol Filter Logs. Only FTP traffic is allowed to access the network. 
Expected Result	EdgeIPS has learned network traffic and created a security policy rule that allows the FTP access pass through.

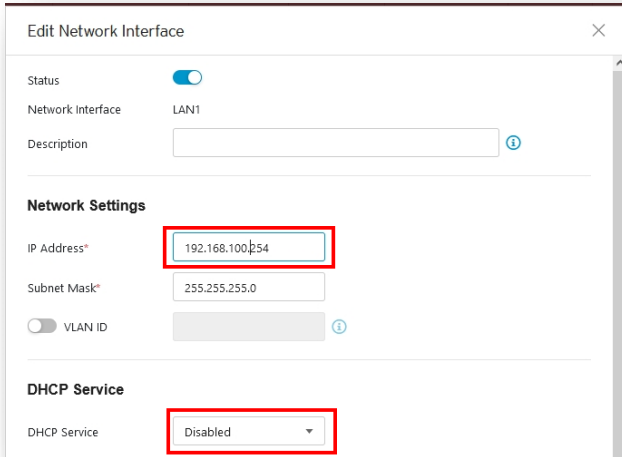
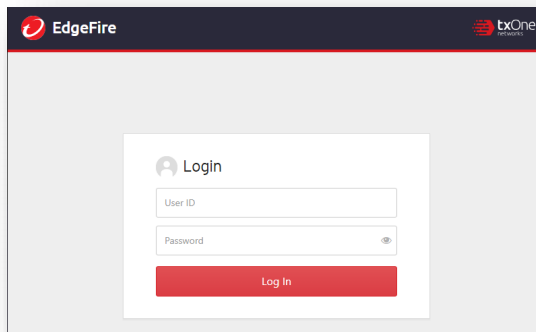
Lab9: EdgeFire initialize settings and registration with EdgeOne

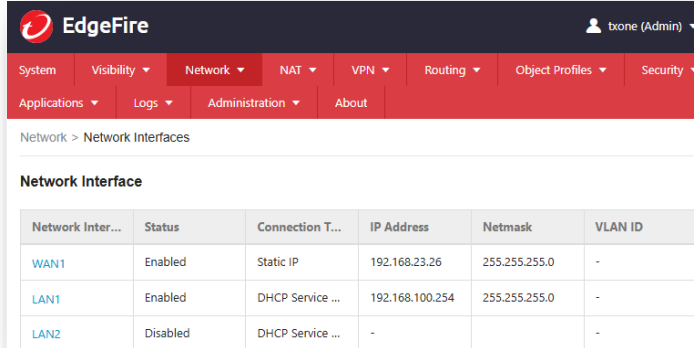
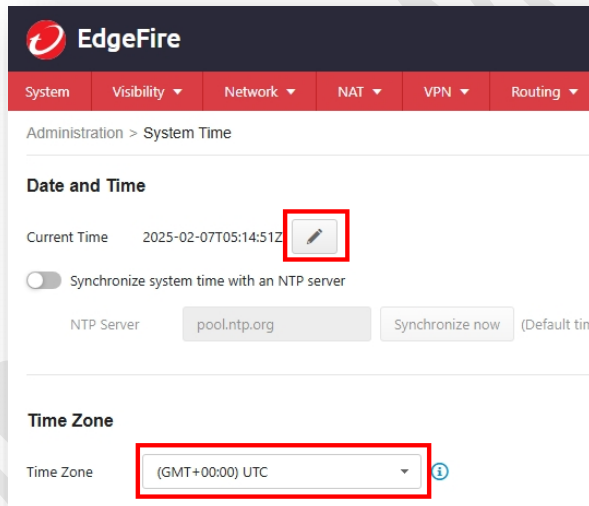
Table 12 EdgeFire Initialize Settings and registration with EdgeOne

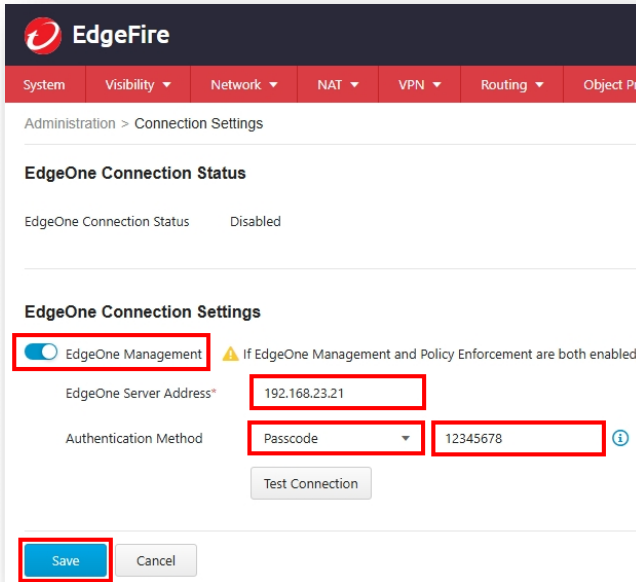
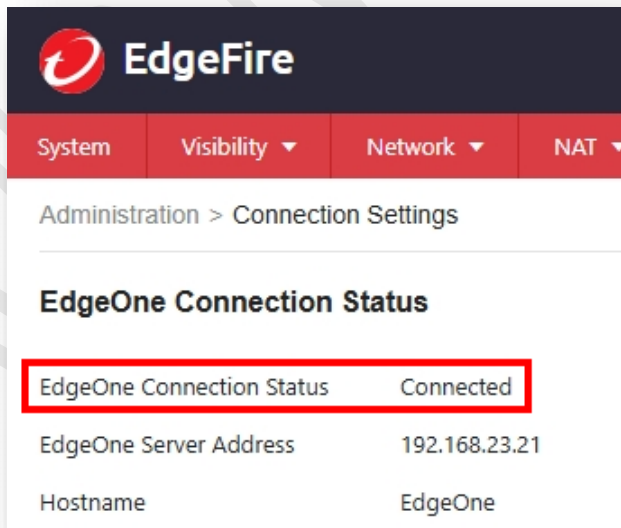
Items	Descriptions
Application target	Host A and B
Steps	The EdgeFire initialize settings and synchronized with EdgeOne SOPs are similar to that for EdgeIPS. Do the Labs 4,5 and 6, but now with EdgeFire. EdgeFire initialization and sync up with EdgeOne - Connect Host A bridged (VMnet11) to any port of LAN1 (From port 1 to port 8) of EdgeFire.  (Host A) Because the VM LabConsole network adapter also binds with a secondary IP address of 192.168.127.1/24, it can directly access any Edge device that is on its default IP address of 192.168.127.254. Edge devices have this default IP address out of the box, or after a factory reset. For the EdgeFire, you need to connect to any of the LAN ports. 

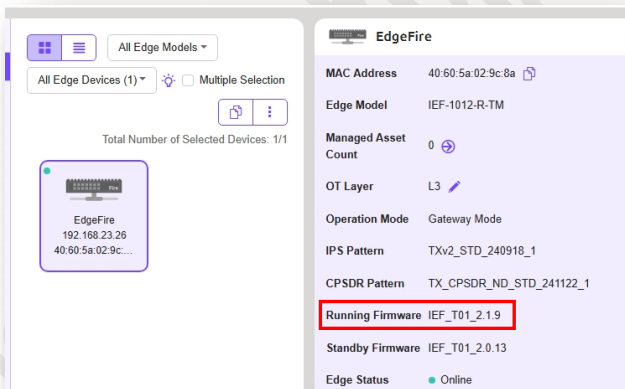
Items	Descriptions
	3. Navigate to https://192.168.127.254 and login with “admin / txone”. 4. Click on the Continue button to accept EULA. Note: A “EULA acceptance” window will show up if the EdgeFire is running firmware v2.  5. Re-configure the administrator account to “txone/P@ssw0rd” 6. Log in to the EdgeFire Device web UI using “txone/P@ssw0rd” 7. Go to Administration > Firmware Management, verify the firmware version of Running partition is v2.0.x or not. If the firmware version is older than v2.1.x, please follow the EdgeIPS Lab 4 to upgrade the firmware to version v2.1.x or above before continuing this lab. 

Items	Descriptions
	8. Go to Administration > System Management 9. Enable "Allow Access from WAN"  10. Go to Network > Network Interface to configure a new IP address for EdgeFire. WAN1 IP Address: 192.168.23.26 Hint: Configure the WAN1 interface first, then configure the LAN1 IP address 

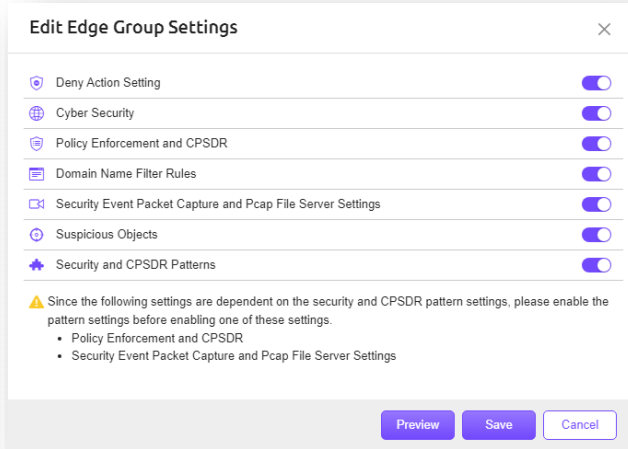
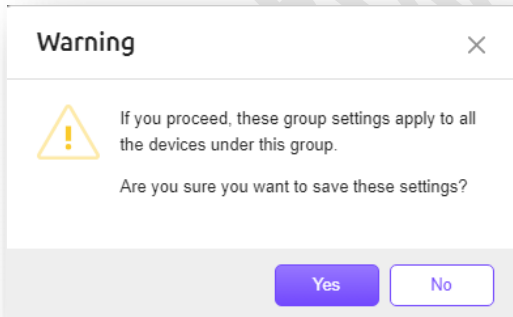
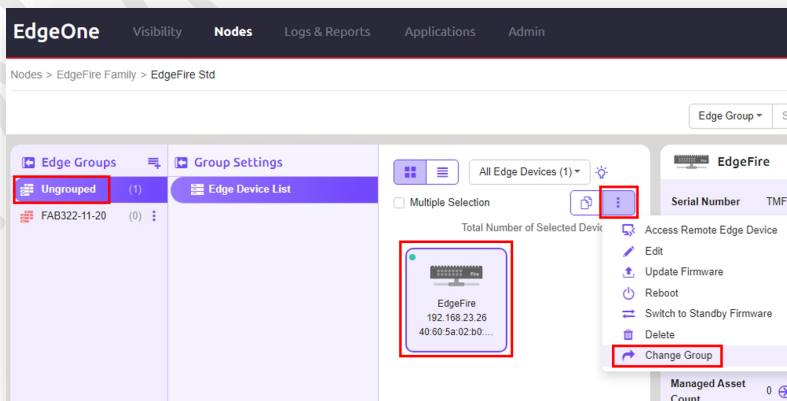
Items	Descriptions
	11. LAN1 IP Address: 192.168.100.254 / 255.255.255.0  12. The EdgeFire web UI will be disconnected after the LAN1 IP address has been configured to 192.168.100.254. Connect the Host A bridged (VMnet11) network cable to the WAN1 port of EdgeFire.  13. Navigate to the EdgeFire web UI: "https://192.168.23.26" and login using user "txone" with password "P@ssw0rd" 

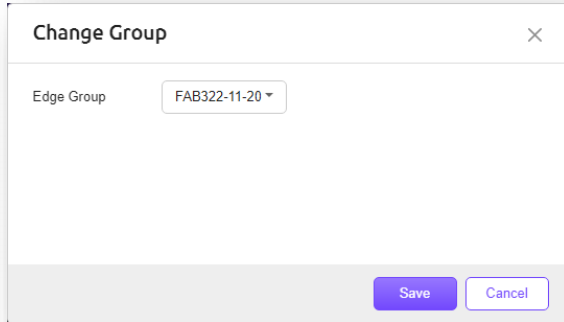
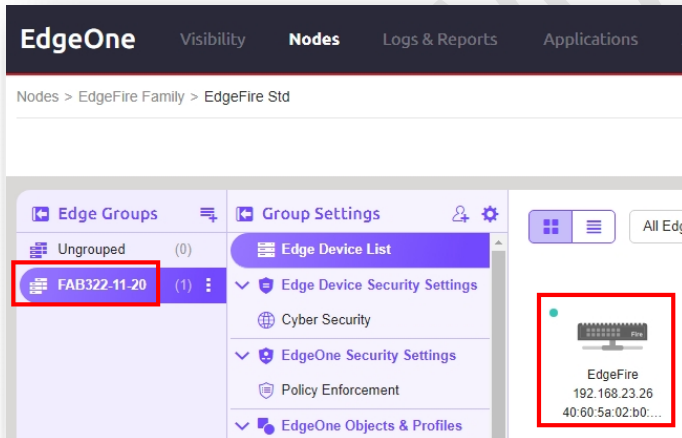
Items	Descriptions
	14. Go to Network > Network Interface and check the Network interface settings.  15. Go to Administration > System Time to configure the time zone to UTC  16. Go to Administration > Connection Settings to configure EdgeOne Sync Settings. 17. Enable EdgeOne Management 18. Set EdgeOne server address to 192.168.23.21.

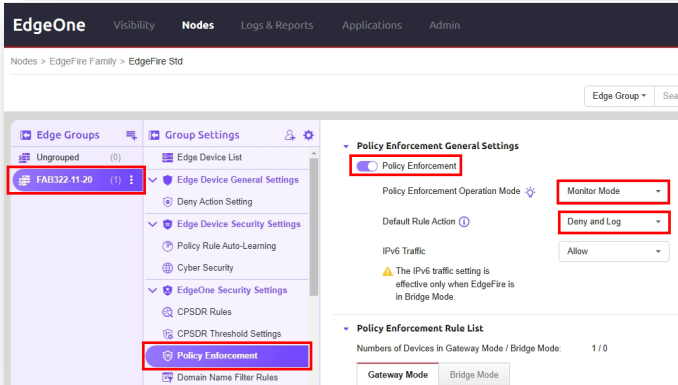
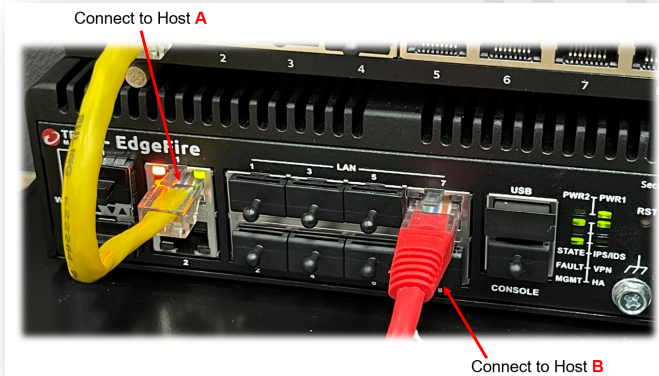
Items	Descriptions
	19. Set the Authentication Method to Passcode, and the password to 12345678  20. Wait for a few seconds. The EdgeOne sync status should now become “Connected”. 

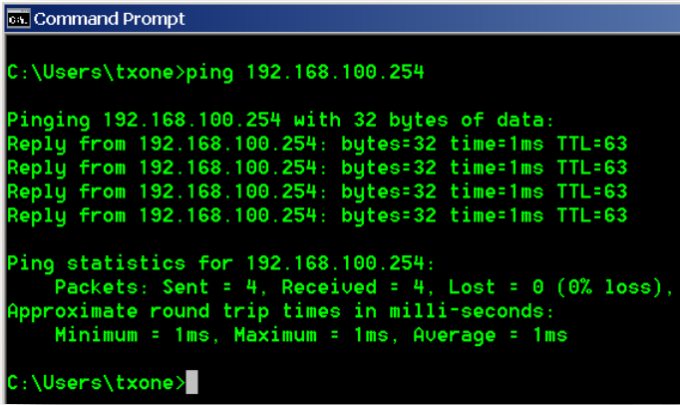
Items	Descriptions
	EdgeFire moved into an EdgeOne group 21. Go to the EdgeOne Web UI at https://192.168.23.21 22. Log with credentials: txone / P@ssw0rd 23. Go to Nodes > EdgeFireFamily > EdgeFire Std. There should be a new EdgeFires in the Edge Group “Ungrouped”. It is same as we did in the EdgeIPS lab. Because the current firmware version of EdgeFire is v2.0.x. It needs to be upgraded to the firmware v2.1 or above before we can connect it to EdgeOne. Upgrade the EdgeFire firmware to v2.1.x Use EdgeOne to deploy the latest version of the EdgeFire firmware and switch to running partition. The steps are similar for the EdgeIPS lab 6 step3. Verify the running firmware version is v2.1.x before going to next step.  24. Go to Nodes > EdgeFire Family > EdgeFire Std and create a new Edge group

Items	Descriptions
	The screenshot shows the EdgeOne web interface. The breadcrumb trail is 'Nodes > EdgeFire Family > EdgeFire Std'. The 'Edge Groups' tab is active, showing a list with 'Ungrouped (1)'. The 'Group Settings' button is highlighted with a red box. To the right, there is a card for 'EdgeFire' with IP '192.168.23.26' and MAC '40:60:5a:02:b0:...'. 25. Name it "FAB322-11-20" The screenshot shows a 'Create New Edge Group' dialog box. The 'Edge Group*' field contains the text 'FAB322-11-20'. There are 'Save' and 'Cancel' buttons at the bottom right. 26. Click "⚙️" to open the Group Settings The screenshot shows the EdgeOne web interface. The breadcrumb trail is 'Nodes > EdgeFire Family > EdgeFire Std'. The 'Edge Groups' tab is active, showing a list with 'Ungrouped (1)' and 'FAB322-11-20 (0)'. The 'FAB322-11-20' group is highlighted with a red box. The 'Group Settings' button is also highlighted with a red box.

Items	Descriptions
	27. Enable all of Edge Group Settings and click Save.  28. Click Yes to apply the settings immediately.  29. Move the EdgeFire from ungrouped to the Edge Group "Fab322-11-20" 

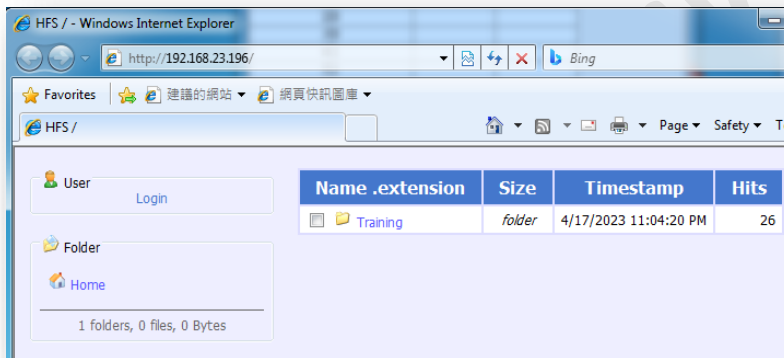
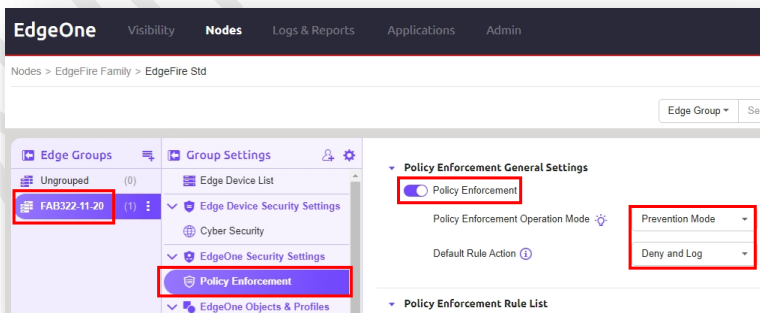
Items	Descriptions
	30. Move the EdgeFire device to the Edge Group “FAB322-11-20”  31. Check the Edge Group “FAB322-11-20” and verify that the new EdgeFire has moved into it.  32. Enable EdgeOne Policy Enforcement and set Policy Enforcement Operation mode to “Monitor Mode”. Remember to click Save to enforce the policy.

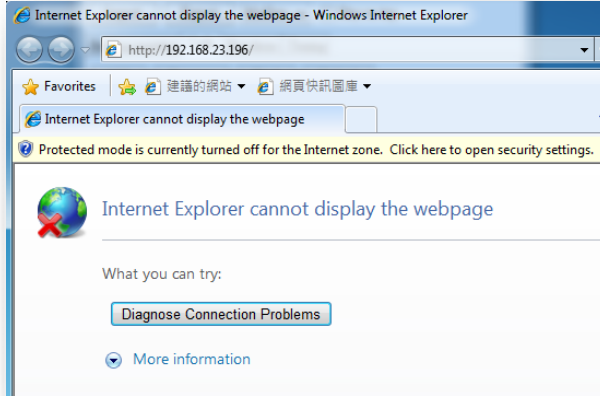
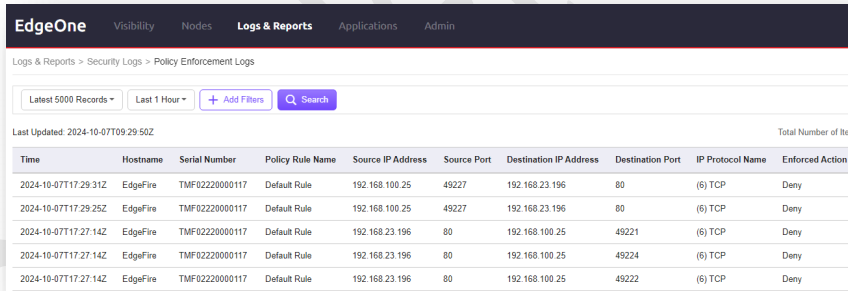
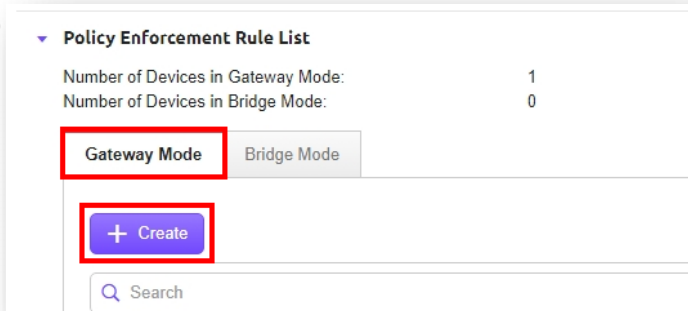
Items	Descriptions
	 33. Connect VM Win7 (Host B) to any LAN1 port of the EdgeFire. Note: By default, all LAN ports (1 to 8) are mapped to LAN1  34. Set the IP address of the Win7 VM to 192.168.100.25/24, and the default gateway point to 192.168.100.254. Also check that the Network adapter is bridged (VMnet11). 35. Open cmd.exe and ping the default gateway 192.168.100.254 to check network connectivity.

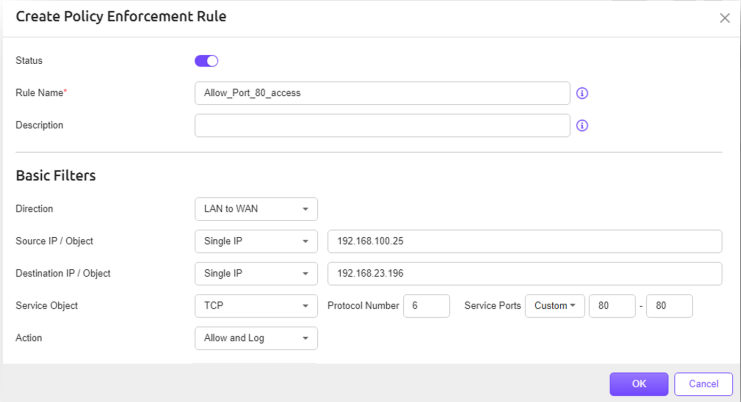
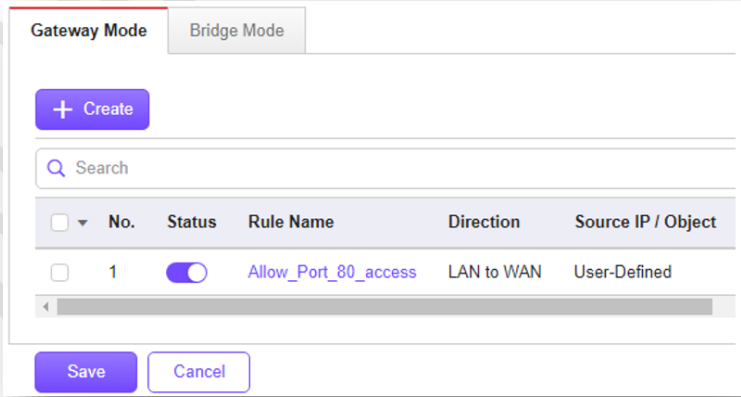
Items	Descriptions
	 <pre> C:\Users\txone>ping 192.168.100.254 Pinging 192.168.100.254 with 32 bytes of data: Reply from 192.168.100.254: bytes=32 time=1ms TTL=63 Reply from 192.168.100.254: bytes=32 time=1ms TTL=63 Reply from 192.168.100.254: bytes=32 time=1ms TTL=63 Reply from 192.168.100.254: bytes=32 time=1ms TTL=63 Ping statistics for 192.168.100.254: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 1ms, Average = 1ms C:\Users\txone> </pre>
Expected Result	EdgeFire initialized and synced up with EdgeOne

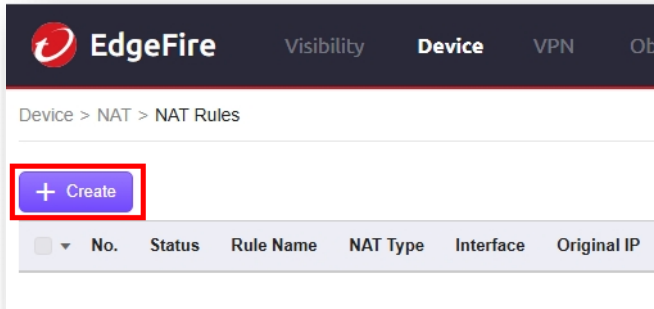
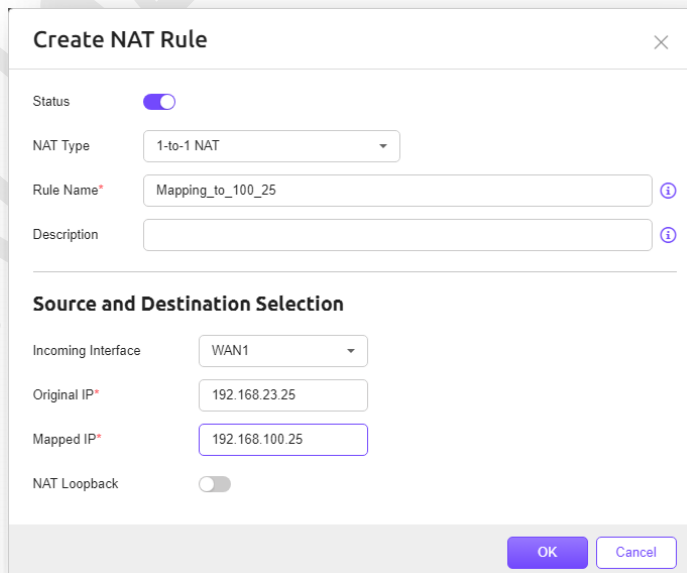
Lab10: EdgeFire NAT Settings

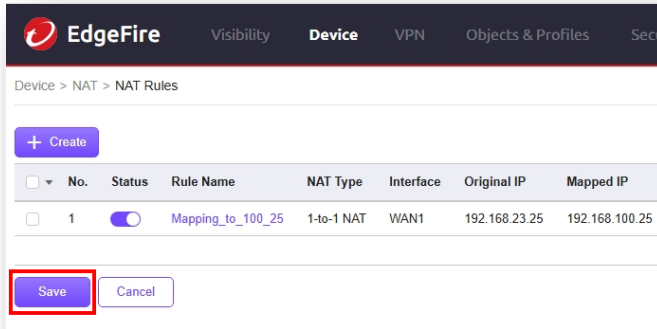
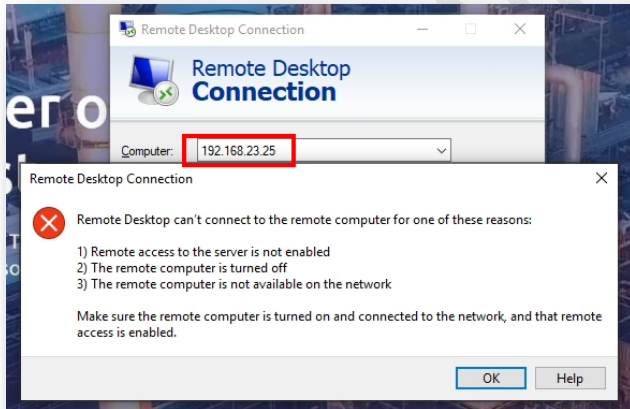
Table 13 EdgeFire NAT Settings

Items	Descriptions
Application target	Host A and B
Steps	(Host B) EdgeFire NAT mode has outbound traffic (LAN1 -> WAN1) as “many to one NAT” configured by default. You can access the HFS web server from VM Win7 (192.168.100.25)  (Host A) Launch the LabConsole VM and go to the EdgeOne web UI. (Host A) Go to Nodes > EdgeFire Family > EdgeFire Std and enable “Prevention Mode”. Remember to click the Save button to enforce policy changes. 

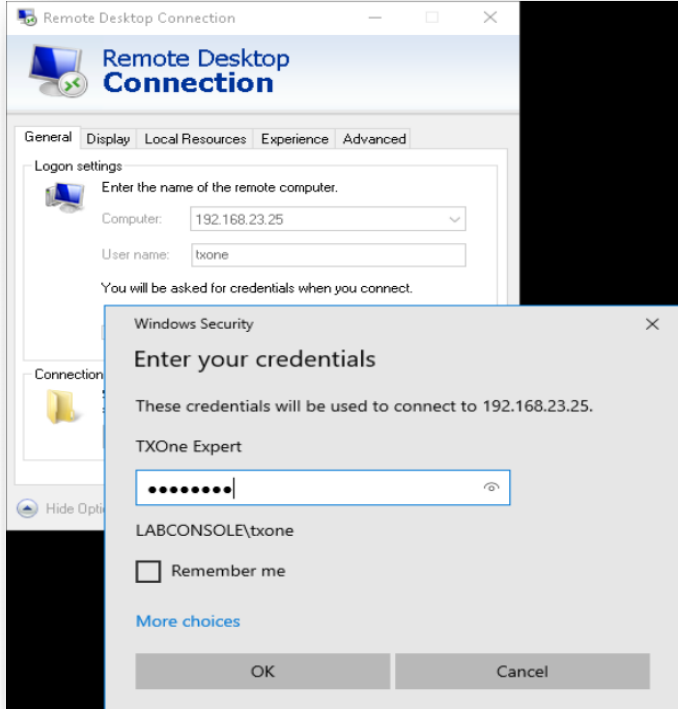
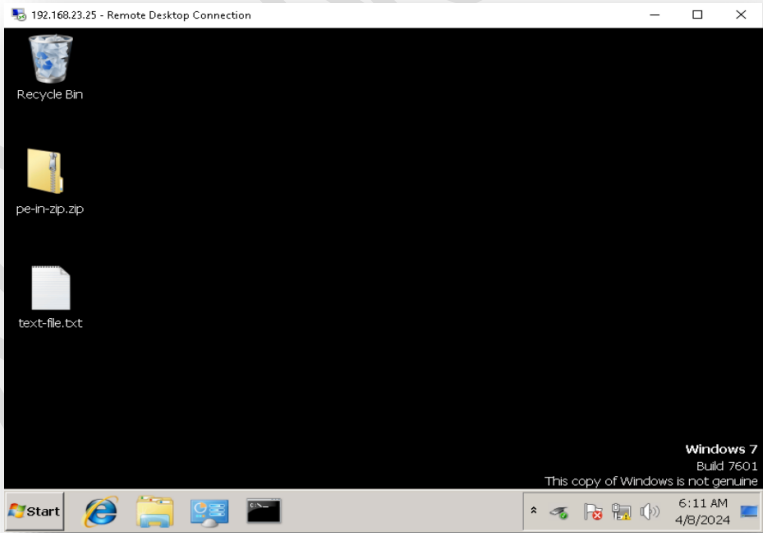
Items	Descriptions
	4. (Host B) Check the connection to the HFS web server again. Now access should be blocked because the default rule action is “Deny and Log”.  5. (Host A) Navigate to the EdgeOne web UI 6. (Host A) Go to Logs & Reports > Security Logs > Policy Enforcement Logs and check the logs.  7. (Host A) Add a Policy Enforcement Rule under the “Gateway Mode” tab. Click on the +Create button. 

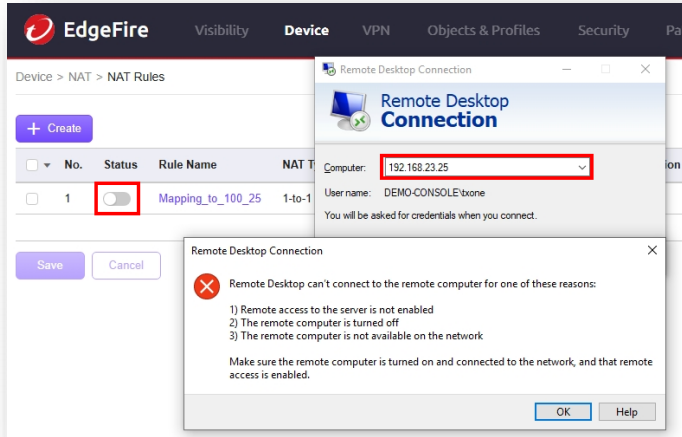
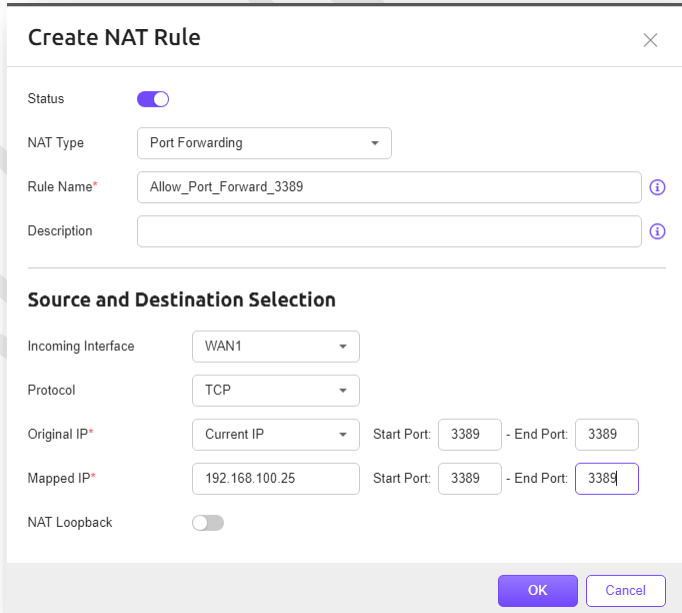
Items	Descriptions														
	8. (Host A) Create a Policy Enforcement Rule and configure a Basic Filter with the following parameters. Then Click OK. <table border="1"> <thead> <tr> <th>Item</th><th>Parameters</th></tr> </thead> <tbody> <tr> <td>Rule Name</td><td>Allow_Port_80_access</td></tr> <tr> <td>Direction</td><td>LAN to WAN</td></tr> <tr> <td>Source IP</td><td>Single IP / 192.168.100.25</td></tr> <tr> <td>Destination IP</td><td>Single IP / 192.168.23.196</td></tr> <tr> <td>Service Object</td><td>TCP, Protocol No. 6, Service Port: Custom, 80 ~ 80</td></tr> <tr> <td>Action</td><td>Accept and Log</td></tr> </tbody> </table>  9. (Host A) Click Save to enforce the policy.  10. (Host B) Check the HFS web server again. Now it is accessible.	Item	Parameters	Rule Name	Allow_Port_80_access	Direction	LAN to WAN	Source IP	Single IP / 192.168.100.25	Destination IP	Single IP / 192.168.23.196	Service Object	TCP, Protocol No. 6, Service Port: Custom, 80 ~ 80	Action	Accept and Log
Item	Parameters														
Rule Name	Allow_Port_80_access														
Direction	LAN to WAN														
Source IP	Single IP / 192.168.100.25														
Destination IP	Single IP / 192.168.23.196														
Service Object	TCP, Protocol No. 6, Service Port: Custom, 80 ~ 80														
Action	Accept and Log														

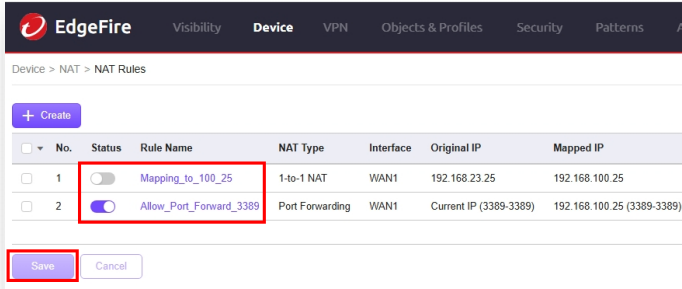
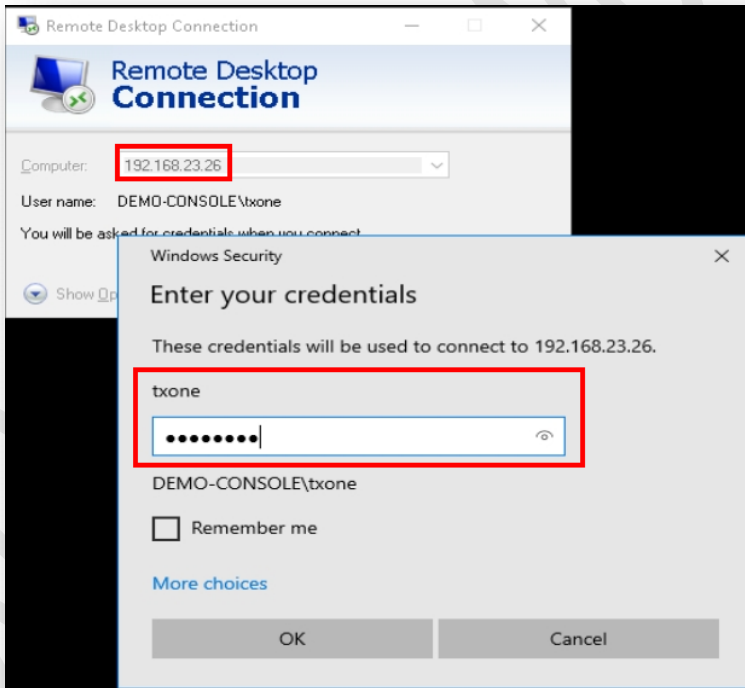
Items	Descriptions												
	Configure 1 to 1 NAT for RDP access from external to internal networks (Host A) Go to the EdgeFire device web UI http://192.168.23.26 (Host A) Go to Device > NAT > NAT Rules and click on the Create button.  (Host A) Create a NAT rule with the following parameters. Click OK. <table border="1"> <thead> <tr> <th>Item</th><th>Parameters</th></tr> </thead> <tbody> <tr> <td>NAT Type</td><td>1 to 1 NAT</td></tr> <tr> <td>Rule Name</td><td>Mapping_to_100_25</td></tr> <tr> <td>Incoming Interface</td><td>WAN1</td></tr> <tr> <td>Original IP</td><td>192.168.23.25</td></tr> <tr> <td>Mapped IP</td><td>192.168.100.25</td></tr> </tbody> </table> 	Item	Parameters	NAT Type	1 to 1 NAT	Rule Name	Mapping_to_100_25	Incoming Interface	WAN1	Original IP	192.168.23.25	Mapped IP	192.168.100.25
Item	Parameters												
NAT Type	1 to 1 NAT												
Rule Name	Mapping_to_100_25												
Incoming Interface	WAN1												
Original IP	192.168.23.25												
Mapped IP	192.168.100.25												

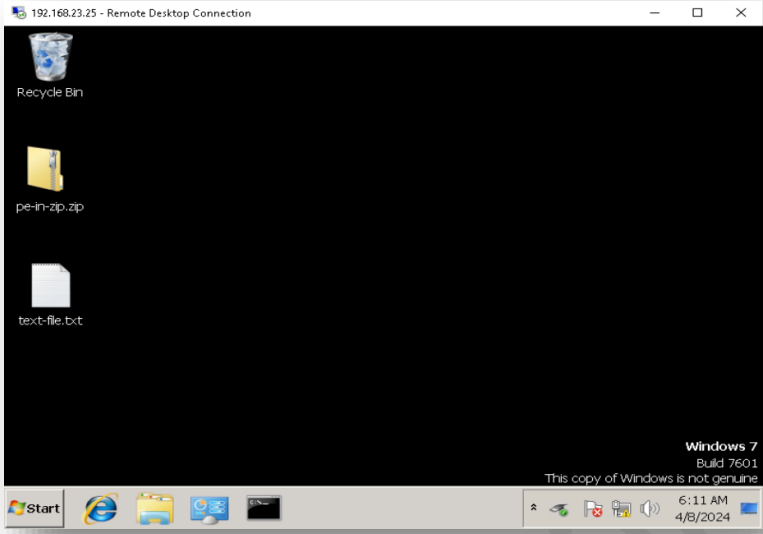
Items	Descriptions														
	4. (Host A) Click Save.  5. (Host A) RDP access from the LabConsole VM to 192.168.23.25. Inbound access to RDP will be blocked by the EdgeFire policy enforcement  6. (Host A) Go to EdgeOne web UI and create a Policy Enforcement Rule and configure a Basic Filter with the following parameters. Then click OK. <table border="1"> <thead> <tr> <th>Item</th><th>Parameters</th></tr> </thead> <tbody> <tr> <td>Rule Name</td><td>Allow_External_RDP_access</td></tr> <tr> <td>Direction</td><td>WAN to LAN</td></tr> <tr> <td>Source IP</td><td>Single IP / 192.168.23.196</td></tr> <tr> <td>Destination IP</td><td>Single IP / 192.168.100.25</td></tr> <tr> <td>Service Object</td><td>TCP, Protocol No. 6, Service Port 3389 ~ 3389</td></tr> <tr> <td>Action</td><td>Accept and Log</td></tr> </tbody> </table>	Item	Parameters	Rule Name	Allow_External_RDP_access	Direction	WAN to LAN	Source IP	Single IP / 192.168.23.196	Destination IP	Single IP / 192.168.100.25	Service Object	TCP, Protocol No. 6, Service Port 3389 ~ 3389	Action	Accept and Log
Item	Parameters														
Rule Name	Allow_External_RDP_access														
Direction	WAN to LAN														
Source IP	Single IP / 192.168.23.196														
Destination IP	Single IP / 192.168.100.25														
Service Object	TCP, Protocol No. 6, Service Port 3389 ~ 3389														
Action	Accept and Log														

Items	Descriptions																		
	Create Policy Enforcement Rule Status ☒ Rule Name* Allow_External_RDP_access ⓘ Description ⓘ Basic Filters Direction WAN to LAN ▾ Source IP / Object Single IP ▾192.168.23.196 Destination IP / Object Single IP ▾192.168.100.25 Service Object TCP ▾ Protocol Number 6 Service Ports Custom ▾3389 - 3389 Action Allow and Log ▾ 7. (Host A) Click Save to enforce the policy. ▼ Policy Enforcement Rule List Number of Devices in Gateway Mode:1 Number of Devices in Bridge Mode:0 Gateway Mode Bridge Mode + Create Q Search <table><tr><th>☐</th><th>No.</th><th>Status</th><th>Rule Name</th><th>Direction</th><th>Source IP / Object</th></tr><tr><td>☐</td><td>1</td><td>☒</td><td>Allow_Port_80_access</td><td>LAN to WAN</td><td>User-Defined</td></tr><tr><td>☐</td><td>2</td><td>☒</td><td>Allow_External_RDP_access</td><td>WAN to LAN</td><td>User-Defined</td></tr></table> Save Cancel 8. (Host A) RDP access from the VM LabConsole to 192.168.23.25 works again. Use the following credentials to Log in. user: txone Password: 12345678	☐	No.	Status	Rule Name	Direction	Source IP / Object	☐	1	☒	Allow_Port_80_access	LAN to WAN	User-Defined	☐	2	☒	Allow_External_RDP_access	WAN to LAN	User-Defined
☐	No.	Status	Rule Name	Direction	Source IP / Object														
☐	1	☒	Allow_Port_80_access	LAN to WAN	User-Defined														
☐	2	☒	Allow_External_RDP_access	WAN to LAN	User-Defined														

Items	Descriptions
	 9. You have now RDP access to the VM Win7  10. (Host A) Go to the EdgeFire Device Web UI and disable 1-to-1 NAT rule "Mapping_to_100_25". 11. (Host A) Go to the internal host via RDP again. RDP access should now be blocked again, even if the policy enforcement rule allows access.

Items	Descriptions														
	 12. (Host A) Create a NAT rule with the following parameters and click OK. <table border="1"> <thead> <tr> <th>Item</th><th>Parameters</th></tr> </thead> <tbody> <tr> <td>NAT Type</td><td>Port Forward</td></tr> <tr> <td>Rule Name</td><td>Allow_Port_Forward_3389</td></tr> <tr> <td>Incoming Interface</td><td>WAN1</td></tr> <tr> <td>Protocol</td><td>TCP</td></tr> <tr> <td>Original IP</td><td>Current IP, Port 3389 ~ 3389</td></tr> <tr> <td>Mapped IP</td><td>192.168.100.25, Port 3389 ~ 3389</td></tr> </tbody> </table> 	Item	Parameters	NAT Type	Port Forward	Rule Name	Allow_Port_Forward_3389	Incoming Interface	WAN1	Protocol	TCP	Original IP	Current IP, Port 3389 ~ 3389	Mapped IP	192.168.100.25, Port 3389 ~ 3389
Item	Parameters														
NAT Type	Port Forward														
Rule Name	Allow_Port_Forward_3389														
Incoming Interface	WAN1														
Protocol	TCP														
Original IP	Current IP, Port 3389 ~ 3389														
Mapped IP	192.168.100.25, Port 3389 ~ 3389														

Items	Descriptions
	13. (Host A) Click Save.  14. (Host A) RDP access to the EdgeFire WAN port (192.168.23.26). The EdgeFire will forward port 3389 to the internal host (VM Win7) successful. 

Items	Descriptions
	
Expected Result	NAT and policy enforcement rules to allow RDP access from external networks to internal networks configured.